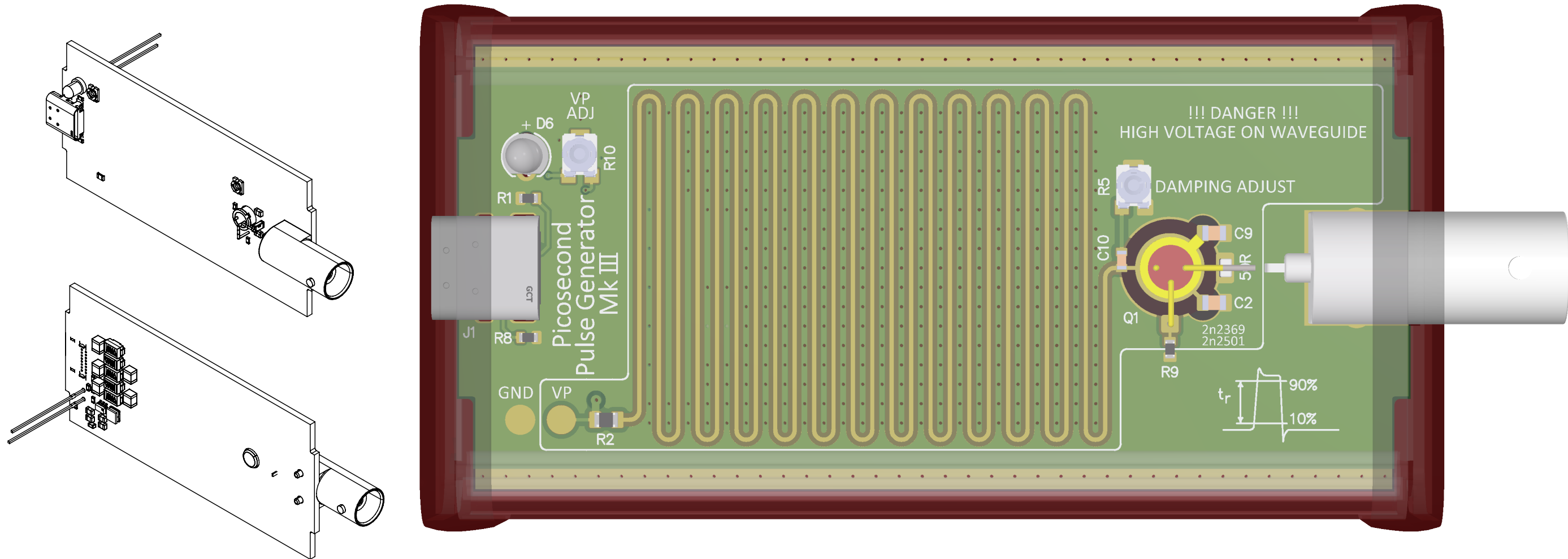
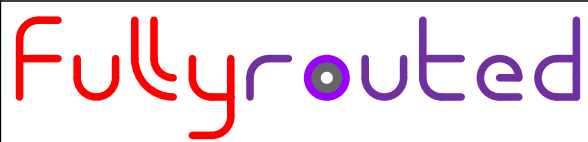


JWpulsegen-V3.PrjPCB

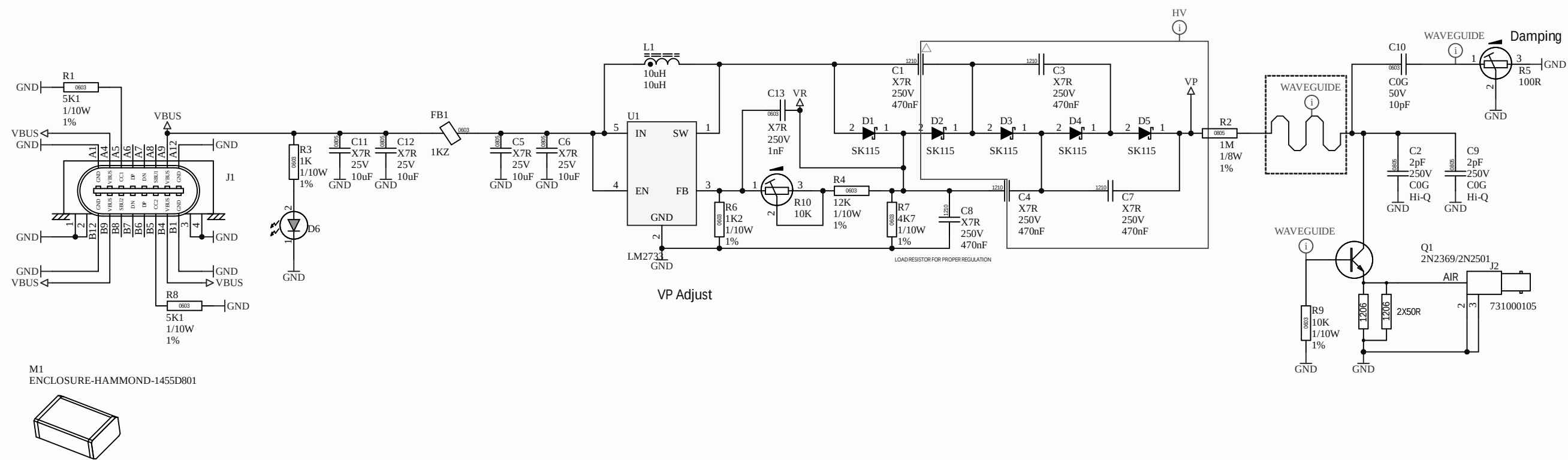


Document Creation Date: 7/8/2025

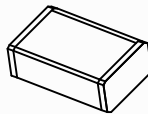
Design : Vincent Himpe



$$VP = VR \times 5$$
$$VR = \frac{1.23V}{1K2} \times (1K2 + R10 + R4)$$
$$C13 : \frac{1}{2 \times 3.14 \times (R4 + R10) \times 8000}$$



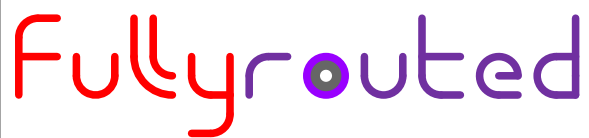
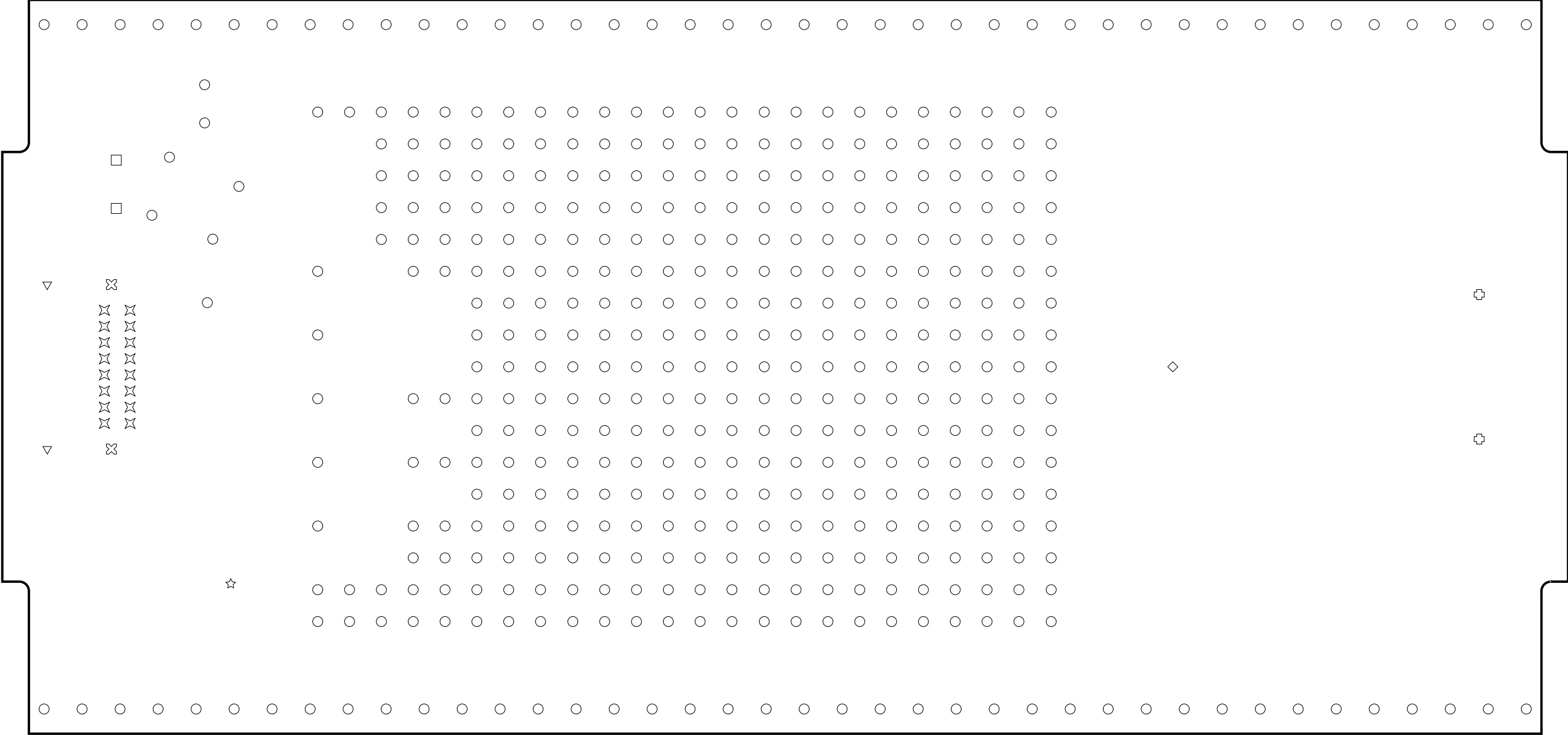
M1
ENCLOSURE-HAMMOND-1455D801



1	2	3	4	5	6
GENERAL					
A	GENERAL				
	1. DO NOT ALTER SUPPLIED COPPER OR DRILL DATA				
	2. NO COPPER BALANCING OR REMOVAL OF UNUSED PADS ALLOWED.				
	3. SILKSCREEN MAY BE CLIPPED / TRIMMED TO EXPOSE COPPER				
	4. PCB DESIGN AND ACCEPTANCE CRITERIA SHALL FOLLOW THE REQUIREMENTS OF IPC-2221, IPC-2222, AND IPC-6012 CLASS 2				
	5. ALL SPECIFICATIONS SHALL BE THE LATEST STANDARDS, UNLESS OTHERWISE NOTED				
B	6. ALL MODIFICATIONS MUST BE COMMUNICATED AND APPROVED IN WRITING.				
	MATERIALS				
	7. MATERIALS SHALL BE ACCORDING TO THE STACKUP DRAWING IN THIS DOCUMENT.				
	8. MATERIAL SHALL HAVE A FLAMABILITY RATING OF UL 94V-0 OR BETTER				
	9. SURFACE FINISH : HASL				
	10. SOLDER MASK COLOR : BLACK				
C	11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm				
	12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL				
	13. THICKNESS LISTED IN LAYER STACK LEGEND REPRESENT FINAL PRESSED VALUES FOR THE PREPREG				
	14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
	QA, ELECTRICAL TEST AND MARKINGS				
D	15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
				Fullyrouted Project JWpulsegen-V3.PrjPCB Version: Variant [No Variations] FABRICATION DRAWING	
1	2	3	4	5	6

DRILL DRAWING

Drill Drawing View (Scale 4:1)



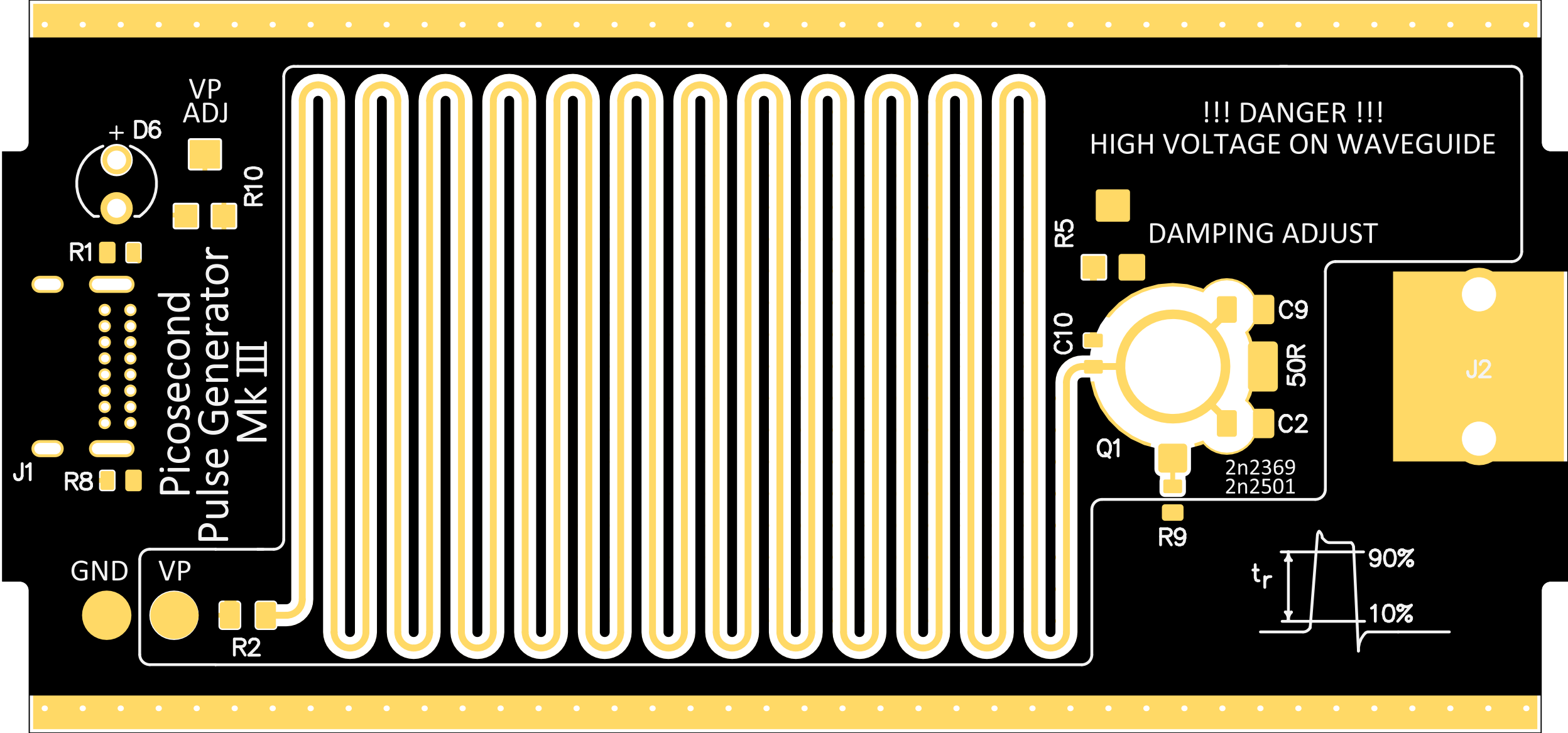
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

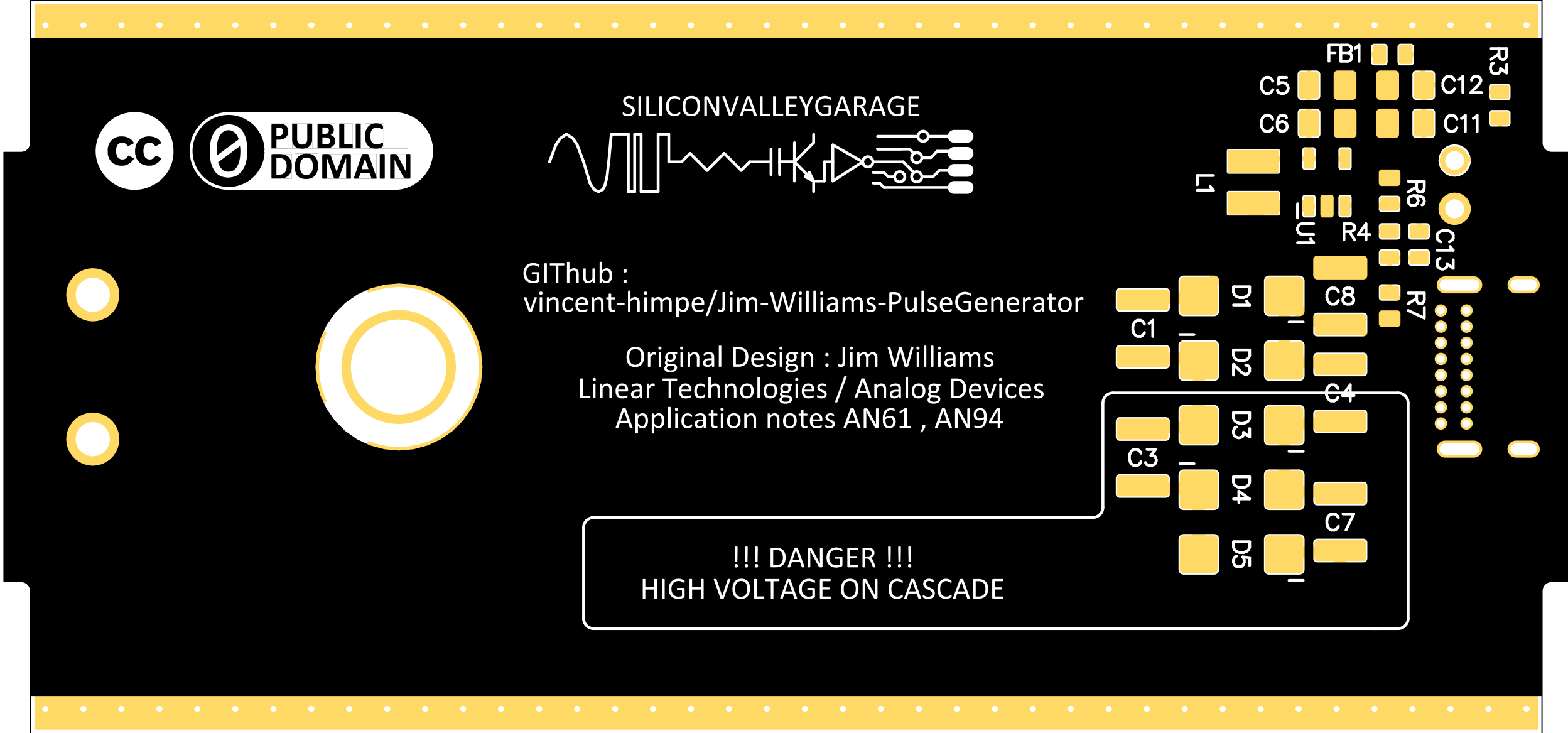
COMPOSITE VIEW FRONT

Top Layer (Scale 4:1)



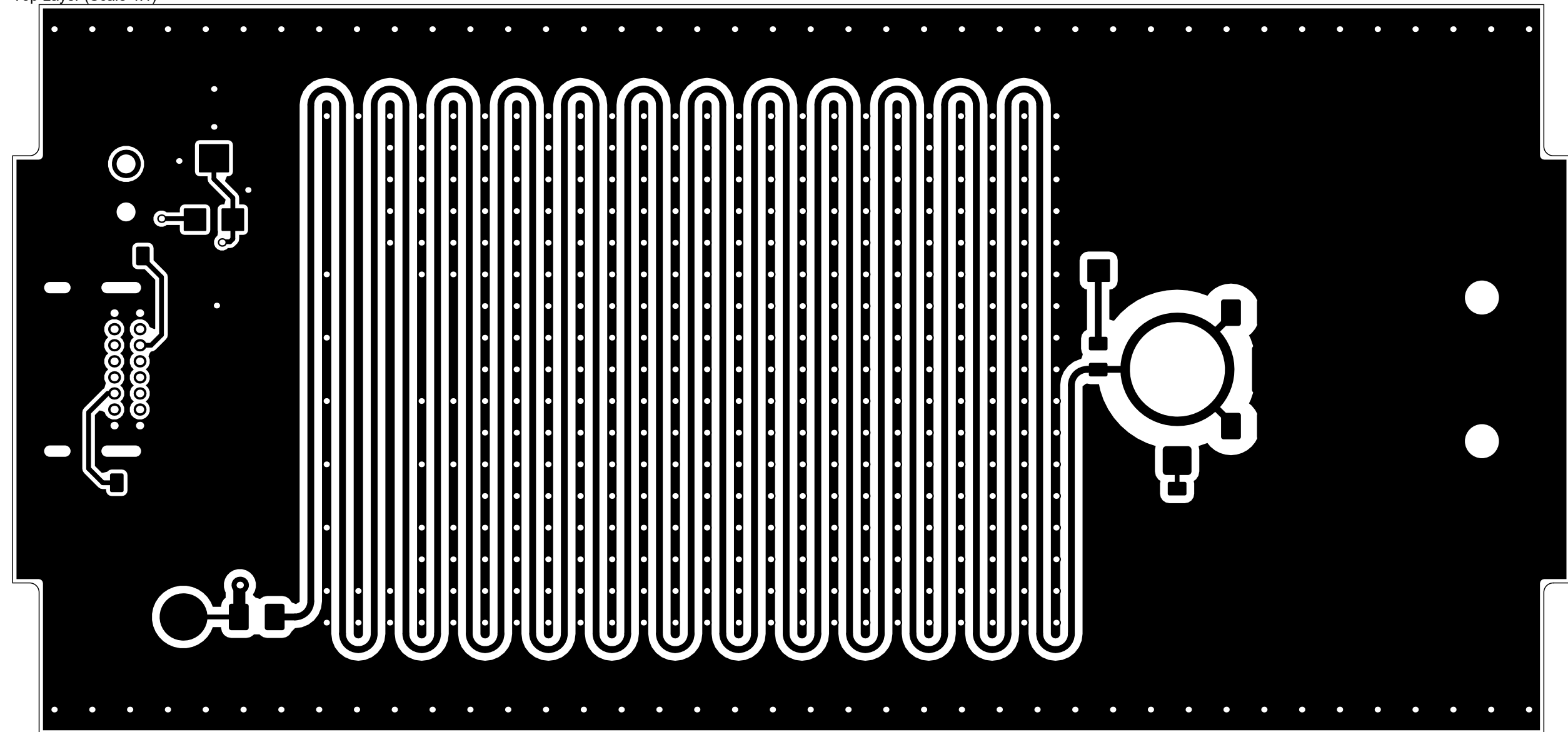
COMPOSITE VIEW BACK

Bottom Layer (Scale 4:1)



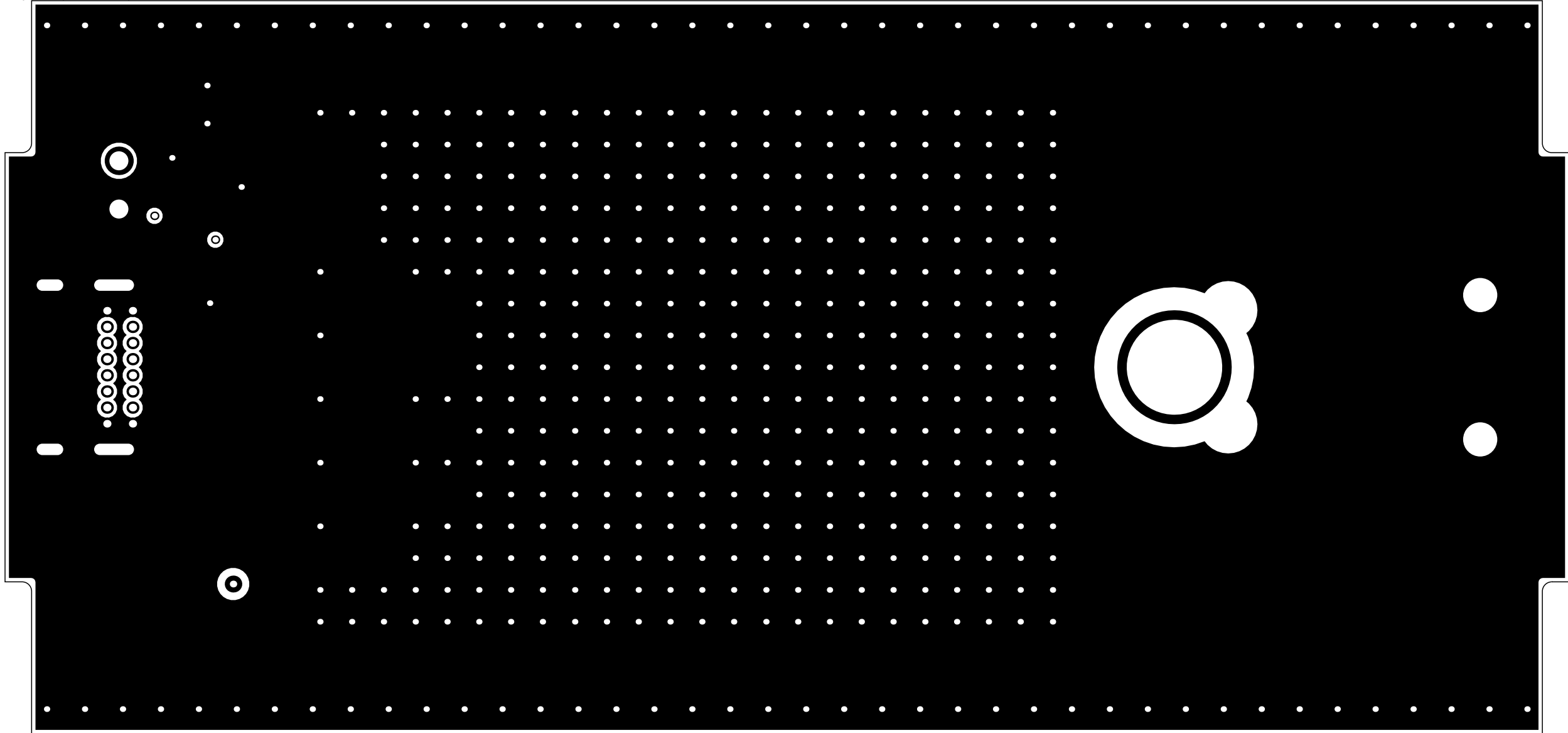
LAYER VIEW : TOP LAYER

Top Layer (Scale 4:1)



LAYER VIEW : MID LAYER 1

Layer 1 (Scale 4:1)



LAYER VIEW : MID LAYER 2

A

A

B

B

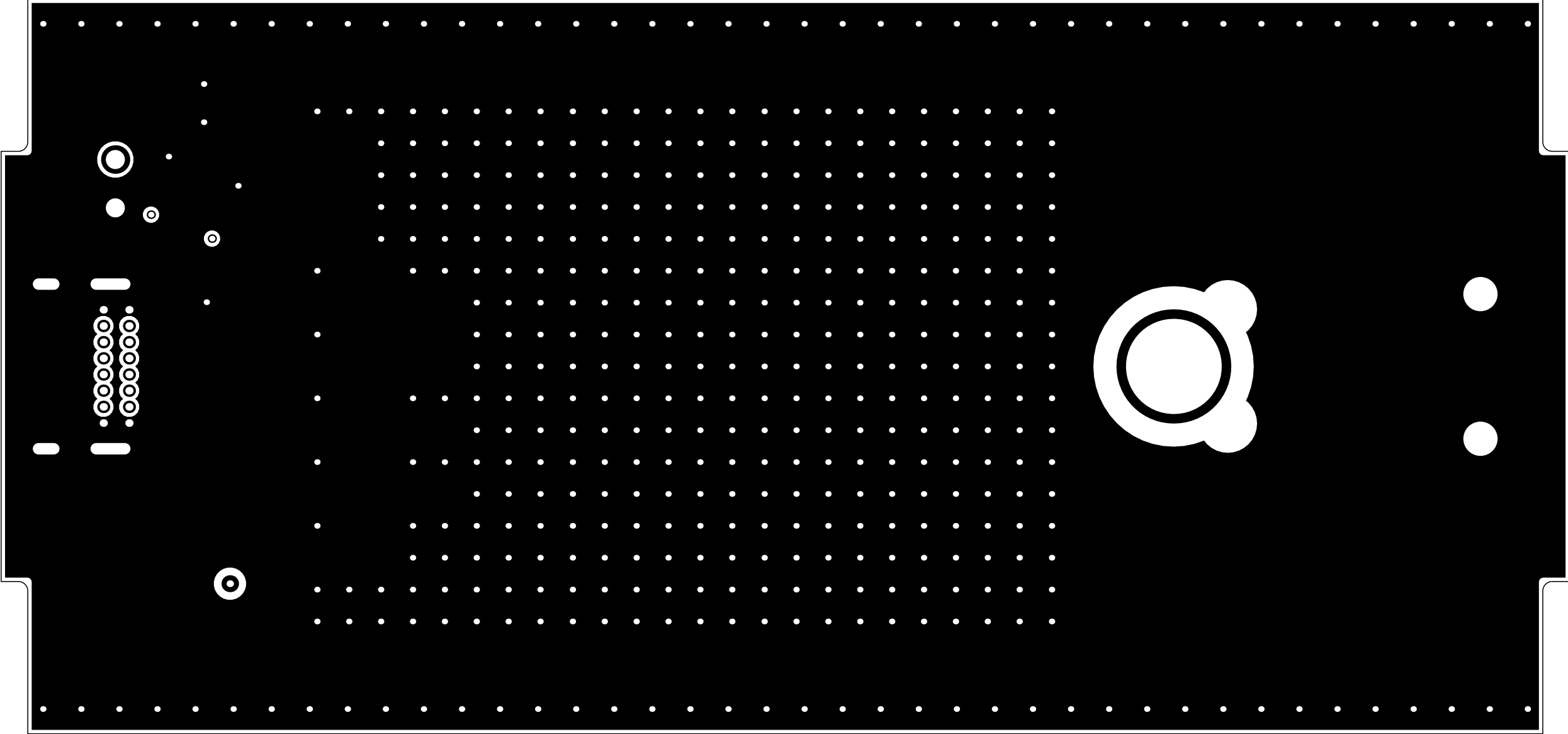
C

C

D

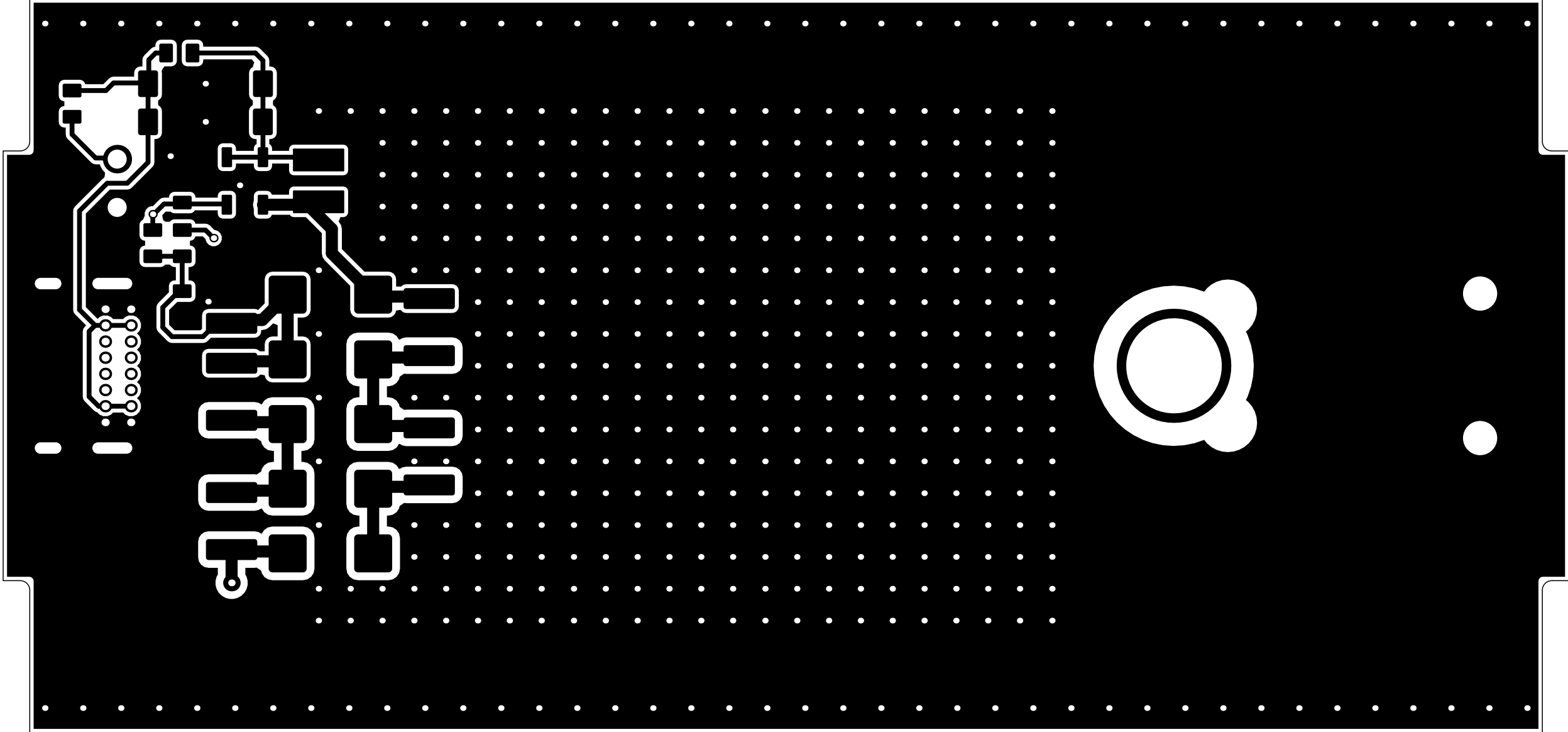
D

Layer 2 (Scale 4:1)



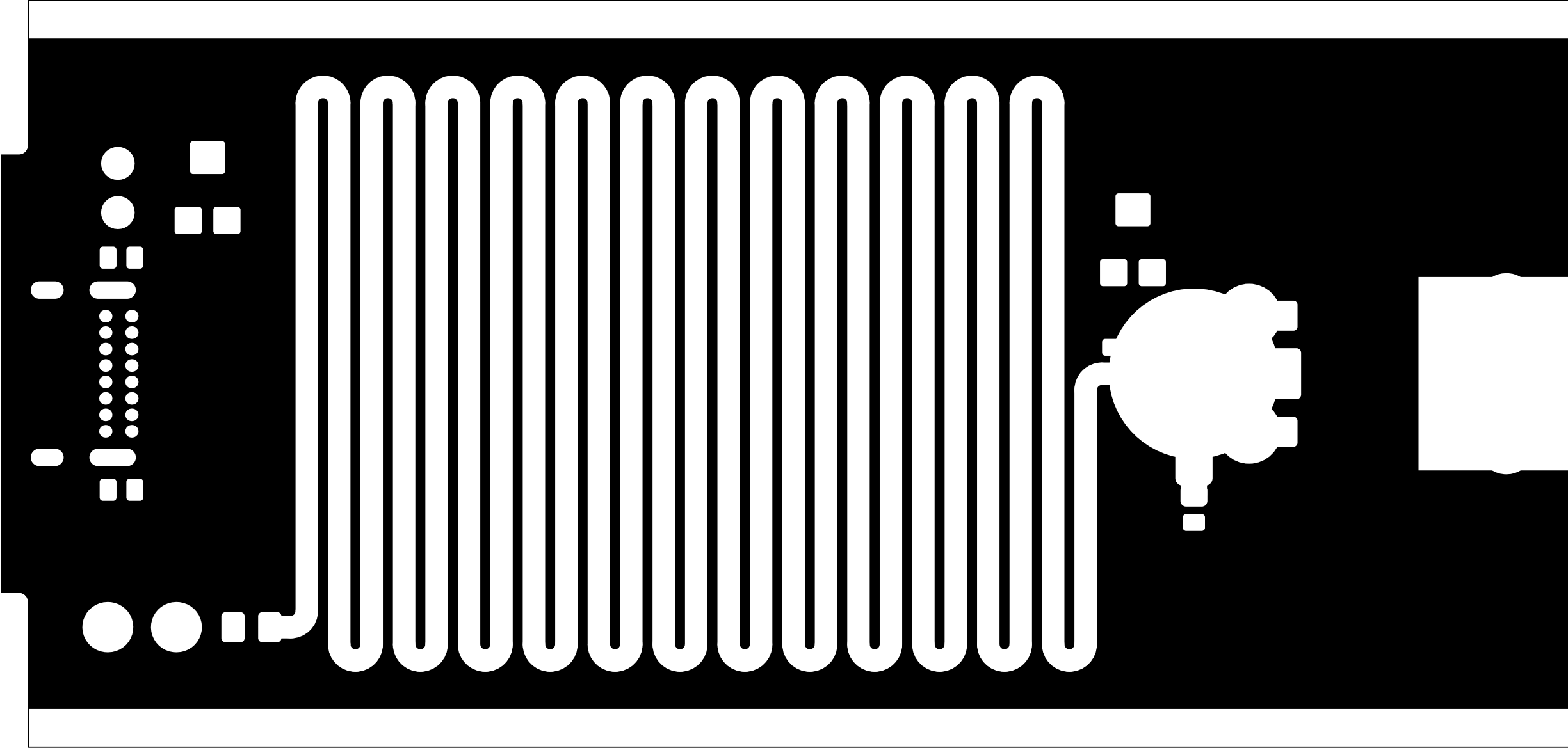
LAYER VIEW : BOTTOM LAYER

Bottom Layer (Scale 4:1)



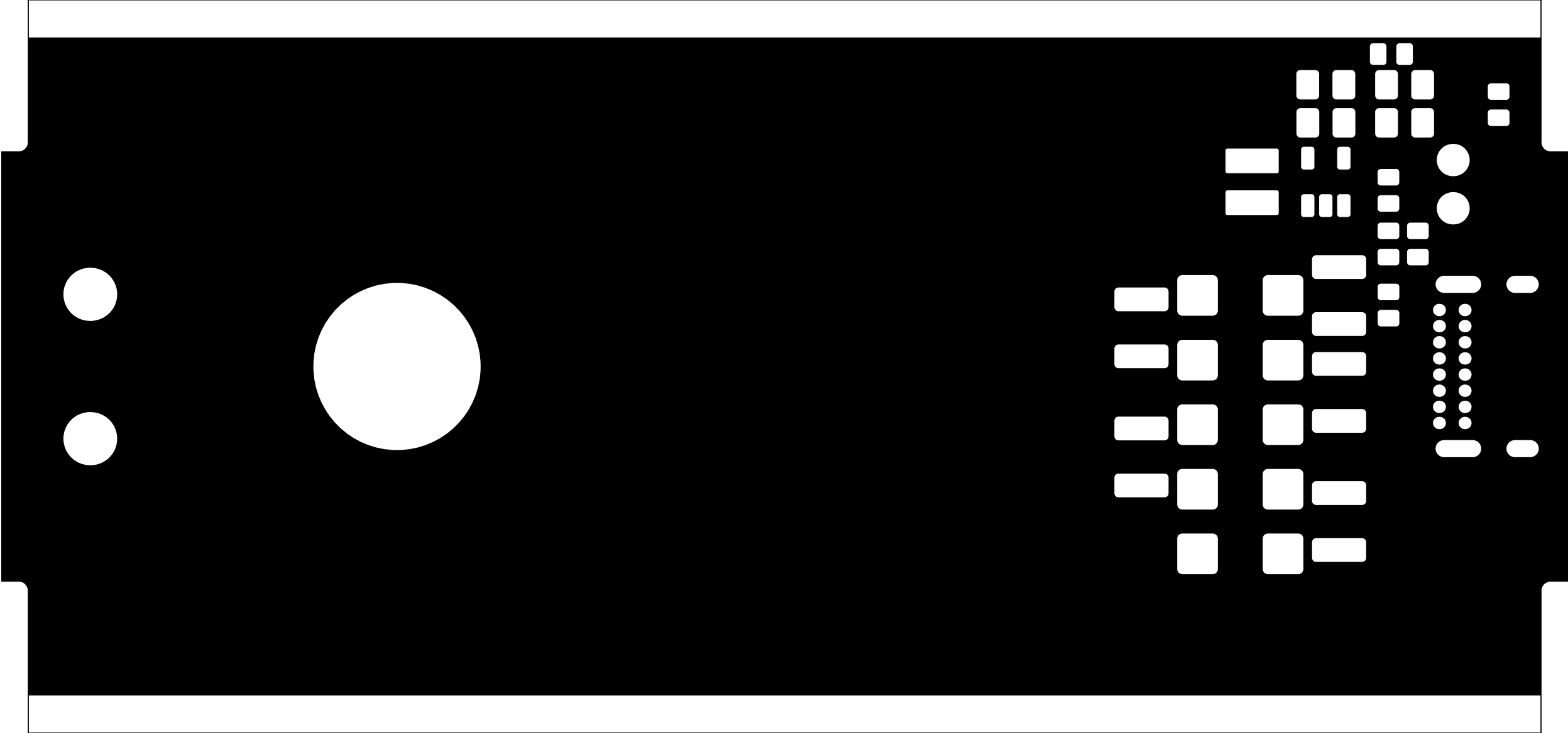
LAYER VIEW : TOP SOLDER MASK

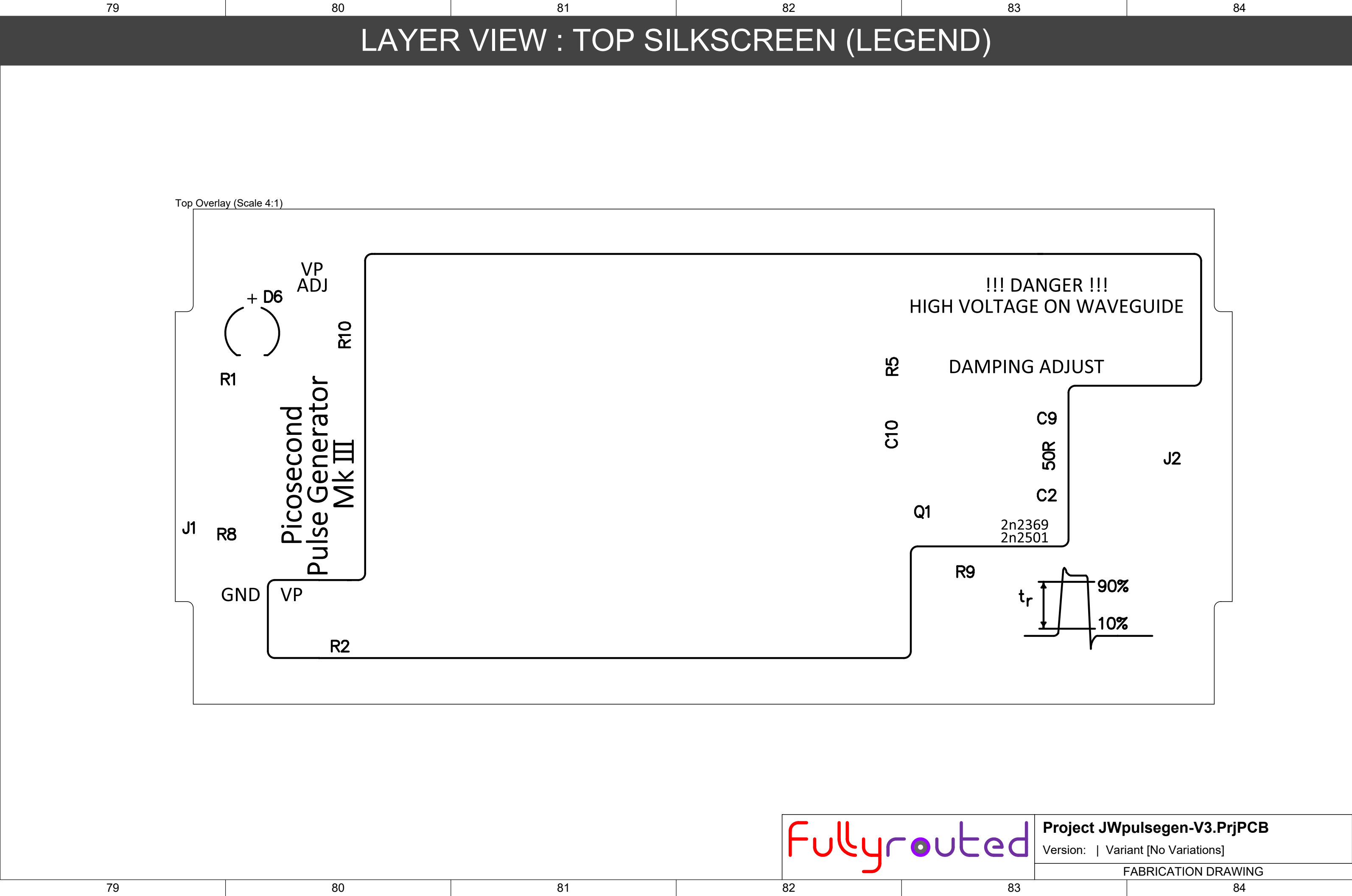
Top Solder (Scale 4:1)



LAYER VIEW : BOTTOM SOLDER MASK

Bottom Solder (Scale 4:1)





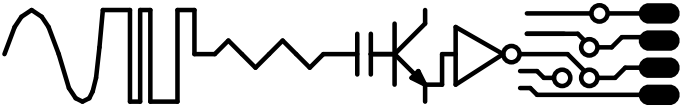
LAYER VIEW : BOTTOM SILKSCREEN (LEGEND)

Bottom Overlay (Scale 4:1)

CC

PUBLIC
DOMAIN

SILICONVALLEYGARAGE



GIThub :

vincent-himpe/Jim-Williams-PulseGenerator

Original Design : Jim Williams

Linear Technologies / Analog Devices

Application notes AN61 , AN94

!!! DANGER !!!

HIGH VOLTAGE ON CASCADE

FB1

C5

C6

L1

C1

C3

D1

D2

D3

D4

D5

C4

C7

C8

R4

R6

R7

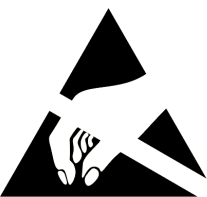
C12

C11

C13

R3

GENERAL



Unless otherwise specified the following rules apply:

- 1. DO NOT DEVIATE FROM ARTWORK OR BOM WITHOUT PRIOR AUTHORIZATION.
- 2. ASSEMBLE AND INSPECT PER IPC-610 CLASS 2

Bill of Materials and Material Handling

- 3. THE BOM CONTAINED IN THIS DOCUMENT IS AS-BUILT. NON-INSTALLED PARTS HAVE BEEN REMOVED. ADDITIONAL BOM FORMATS ARE AVAILABLE IN THE PROJECT FILES
- 4. ANY PART SUBSTITUTIONS MUST BE APPROVED IN WRITING BEFORE ASSEMBLY
- 5. ALL MATERIALS MUST BE PROCURED FROM MANUFACTURER AUTHORIZED DISTRIBUTORS OR THE ORIGINAL MANUFACTURER
- 6. ALL COMPONENTS AND BOARDS TO BE HANDLED AND STORED ACCORDING TO IPC GUIDELINES
- 7. ESD CONTROL PER IPC RULES

Soldering

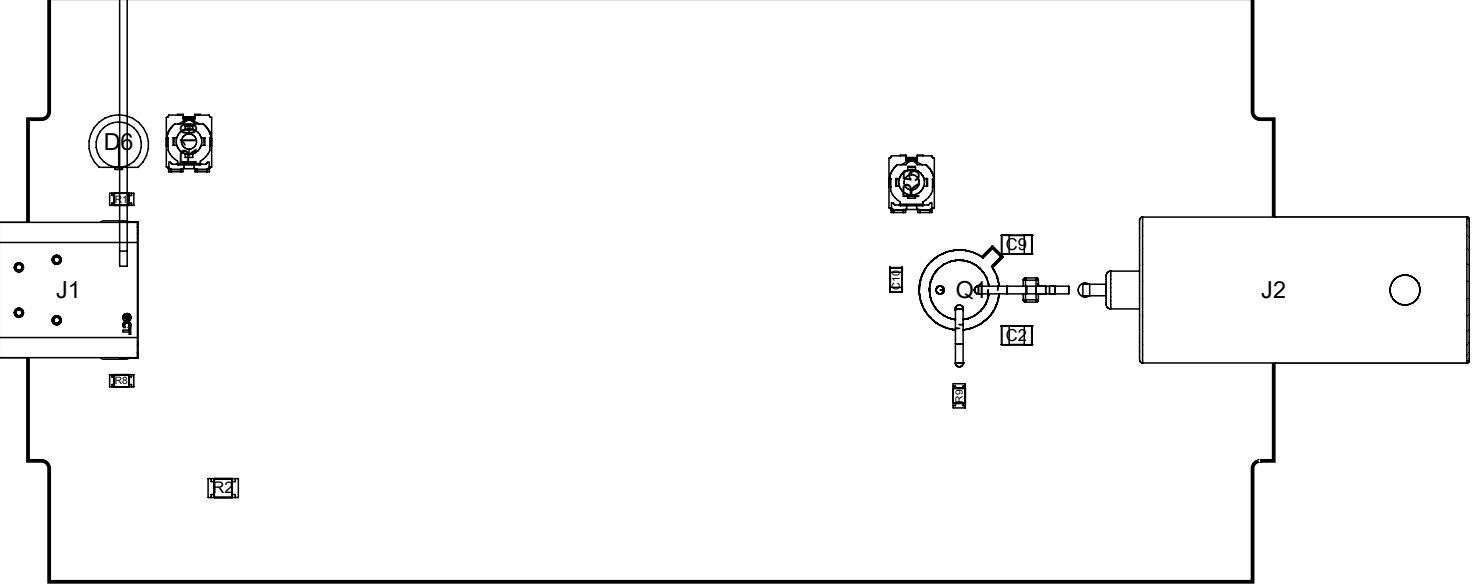
- 8. SOLDERING TO BE DONE USING SN37PB63 ALLOY USING ALLOY MANUFACTURER RECOMMENDED NO-CLEAN FLUX
- 9. BGA COMPONENTS WITH LEAD-FREE CONNECTIONS NEED TO BE REBALLED WITH SN63PB37. MIXING OF ALLOYS IS NOT PERMITTED.
- 10. SOLDERING PREFERRABLY TO BE DONE USING NITROGEN ATMOSPHERE
- 11. SURPLUS COMPONENTS TO VACUUM SEALED WITH DESSICANT IN ANTISTATIC BAGS
- 12. INCOMING MATERIAL (BOARDS AND COMPONENTS) NEEDS TO BE INSPECTED FOR HUMIDITY AND BAKED IF NEEDED PRIOR TO USE.
- 13. MANUAL REWORK / TOUCHUP TO BE DONE USING SAME ALLOY AND APPROPRIATE FLUX. FLUX MUST BE REMOVED.

2D VIEW

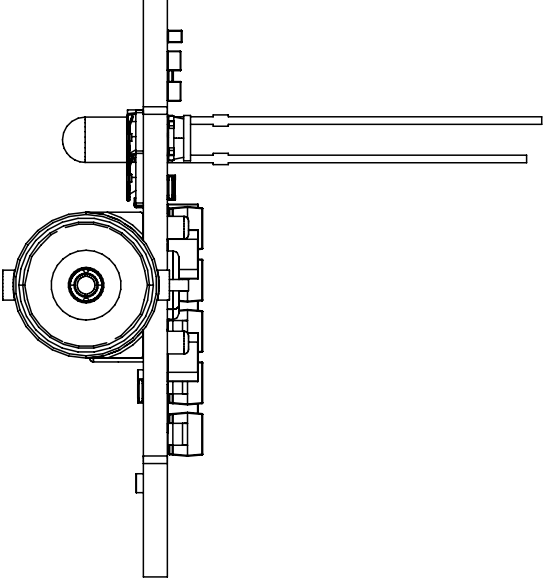
View from Front side (Scale 2:1)



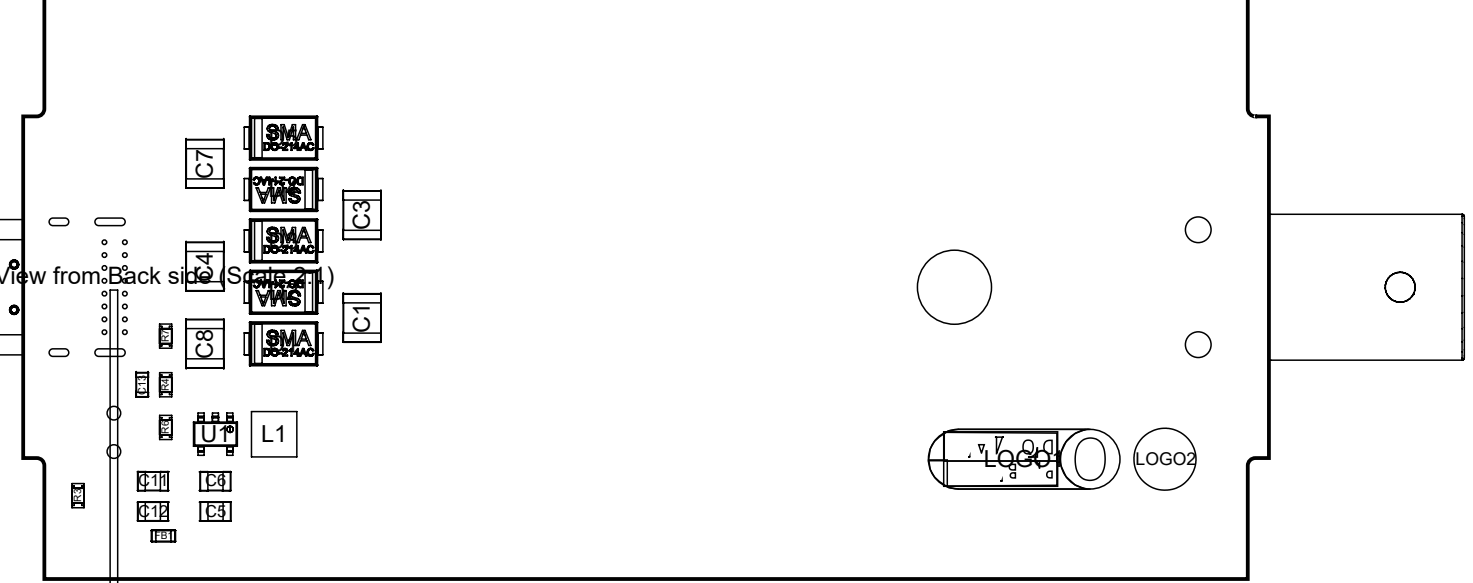
View from Top side (Scale 2:1)



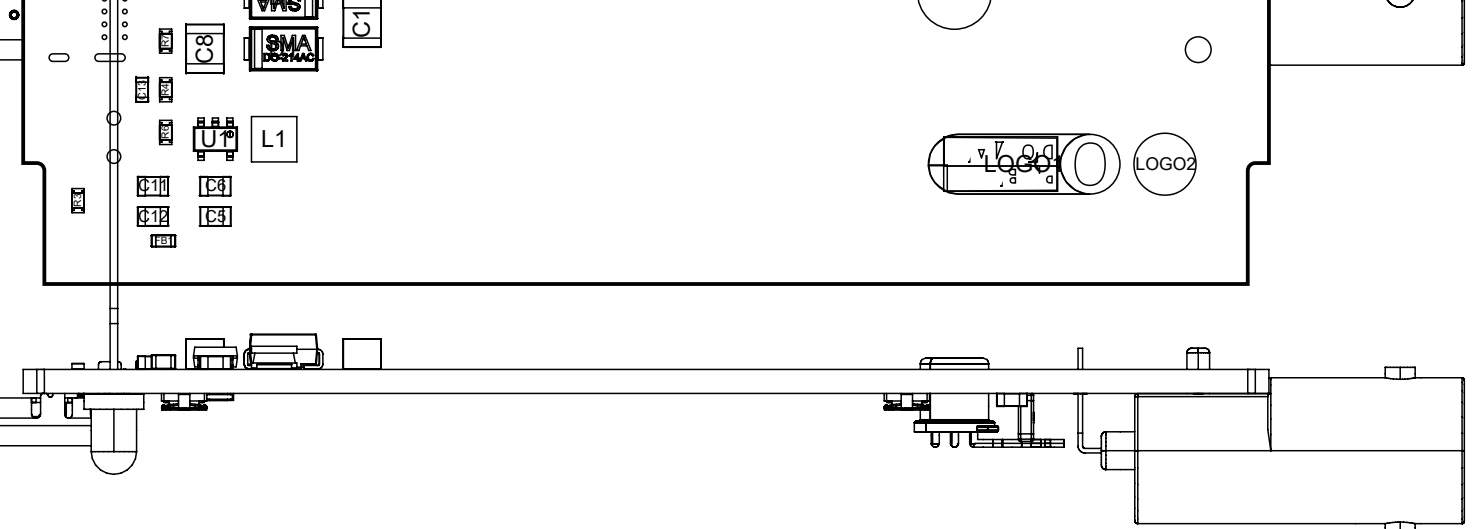
View from Right side (Scale 2:1)



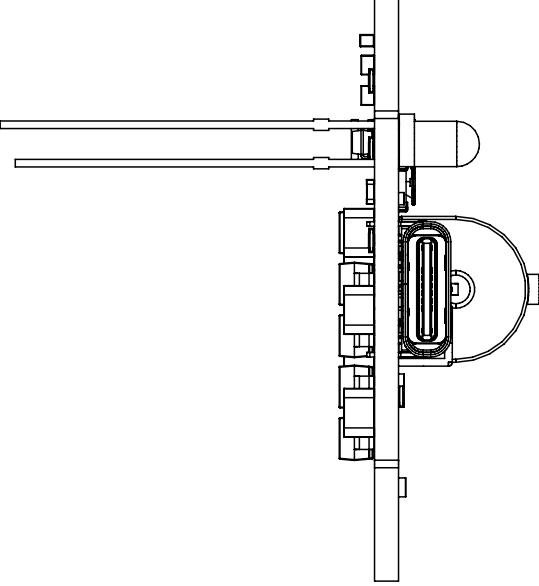
View from Bottom side (Scale 2:1)



View from Back side (Scale 2:1)



View from Left side (Scale 2:1)



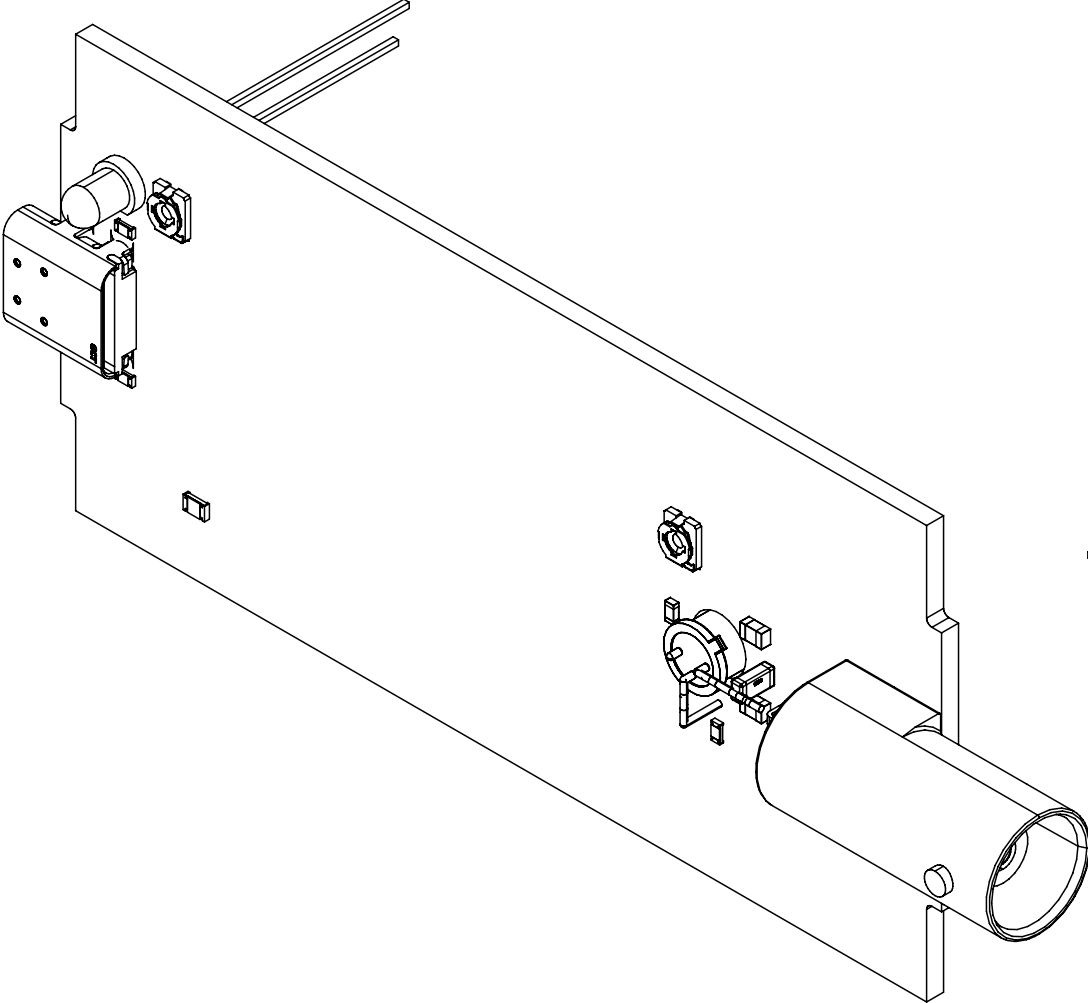
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

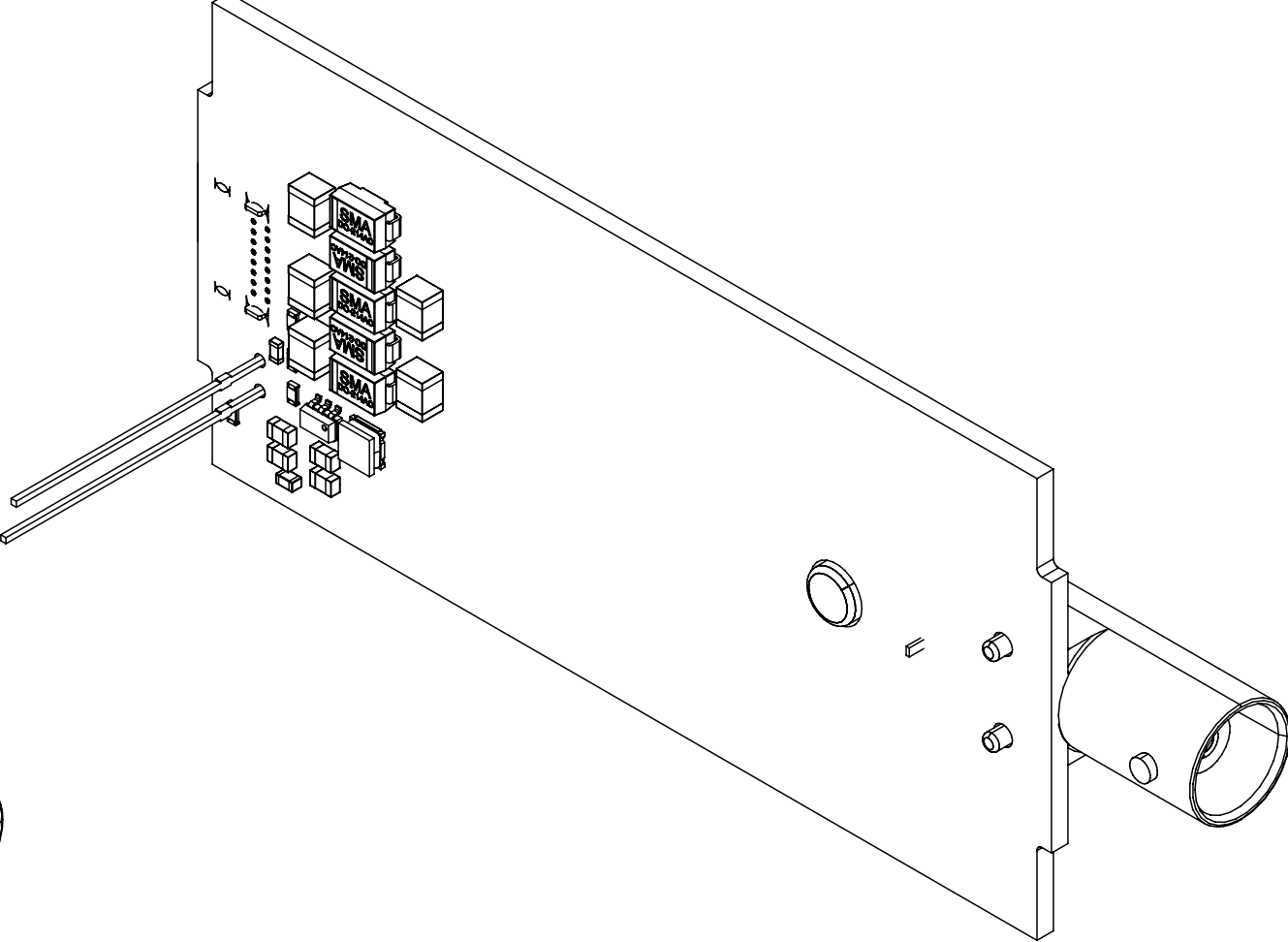
ASSEMBLY DRAWING

3D VIEW

View from Top side (Scale 2:1)

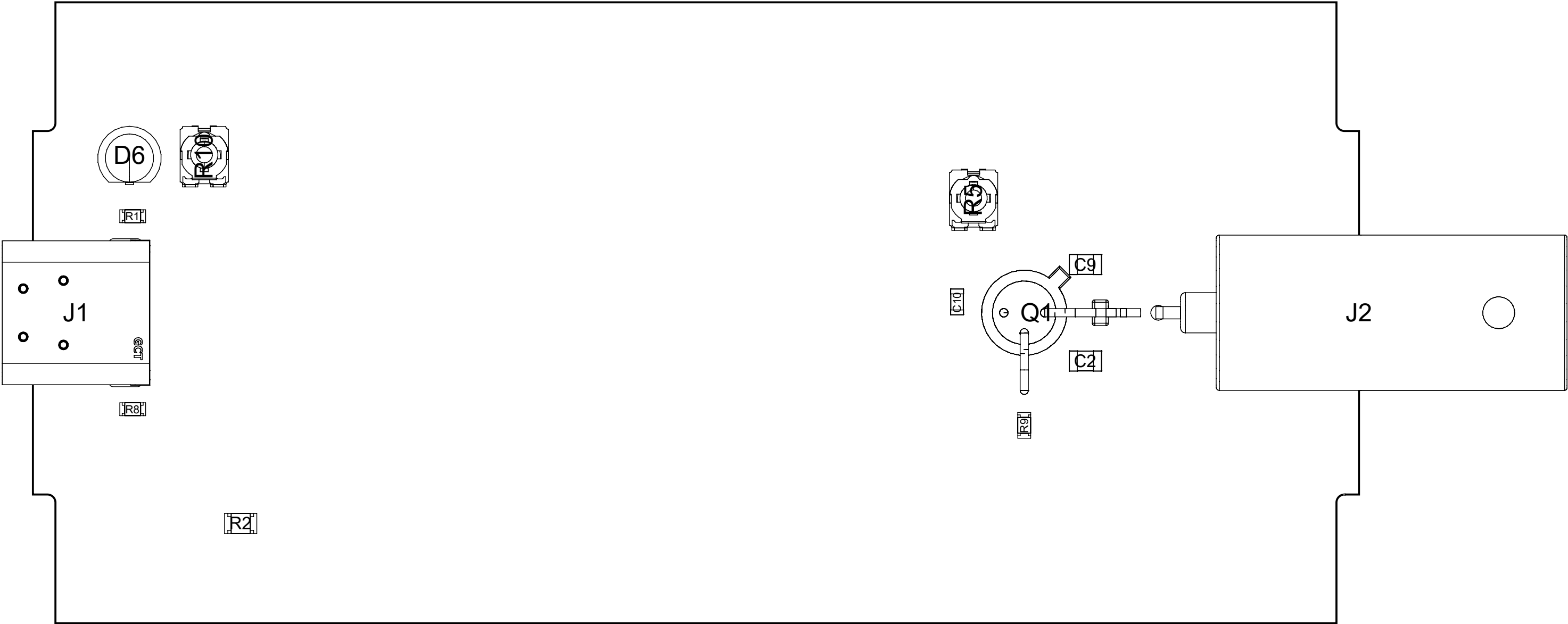


View from Bottom side (Scale 2:1)



1	2	3	4	5	6
Bill Of Materials					

DESIGNATORS FRONT



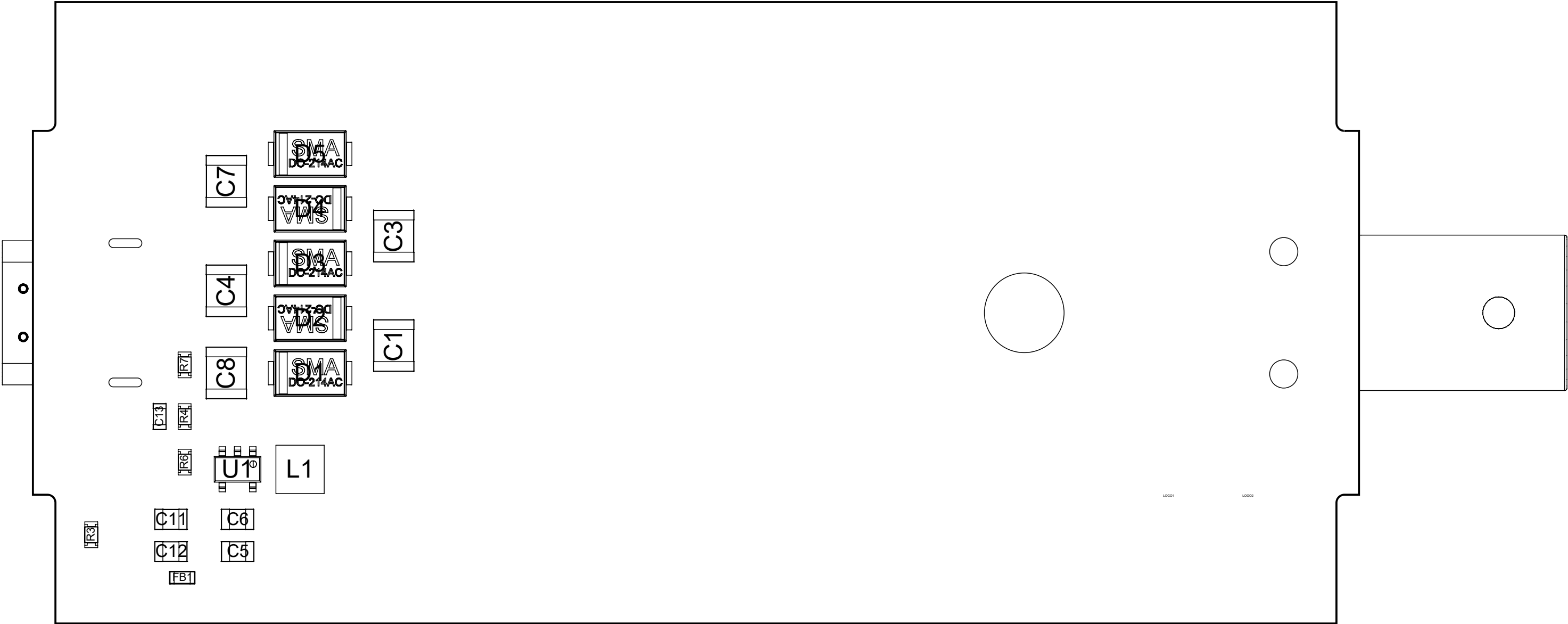
View from Top side (Scale 4:1)



Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

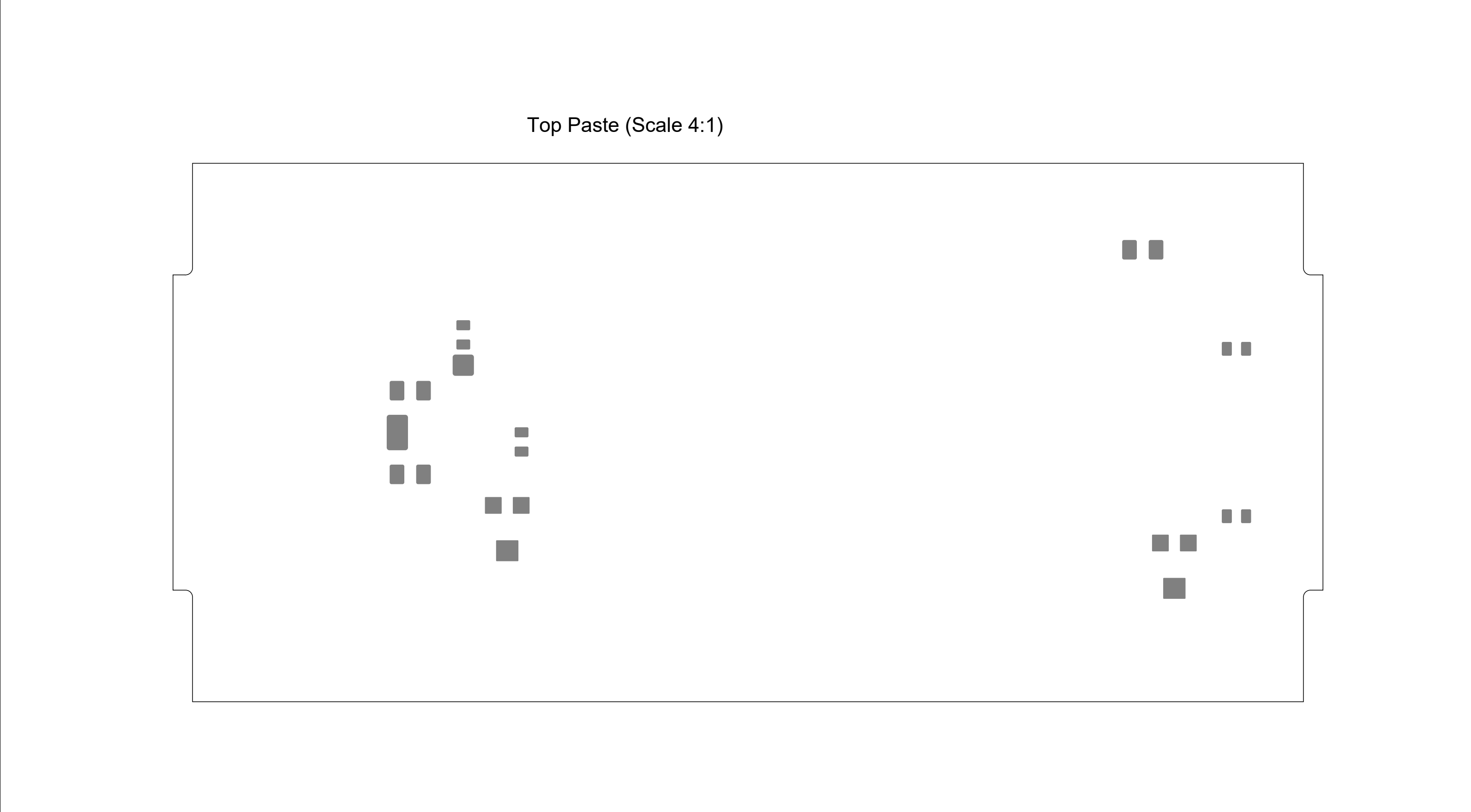
ASSEMBLY DRAWING

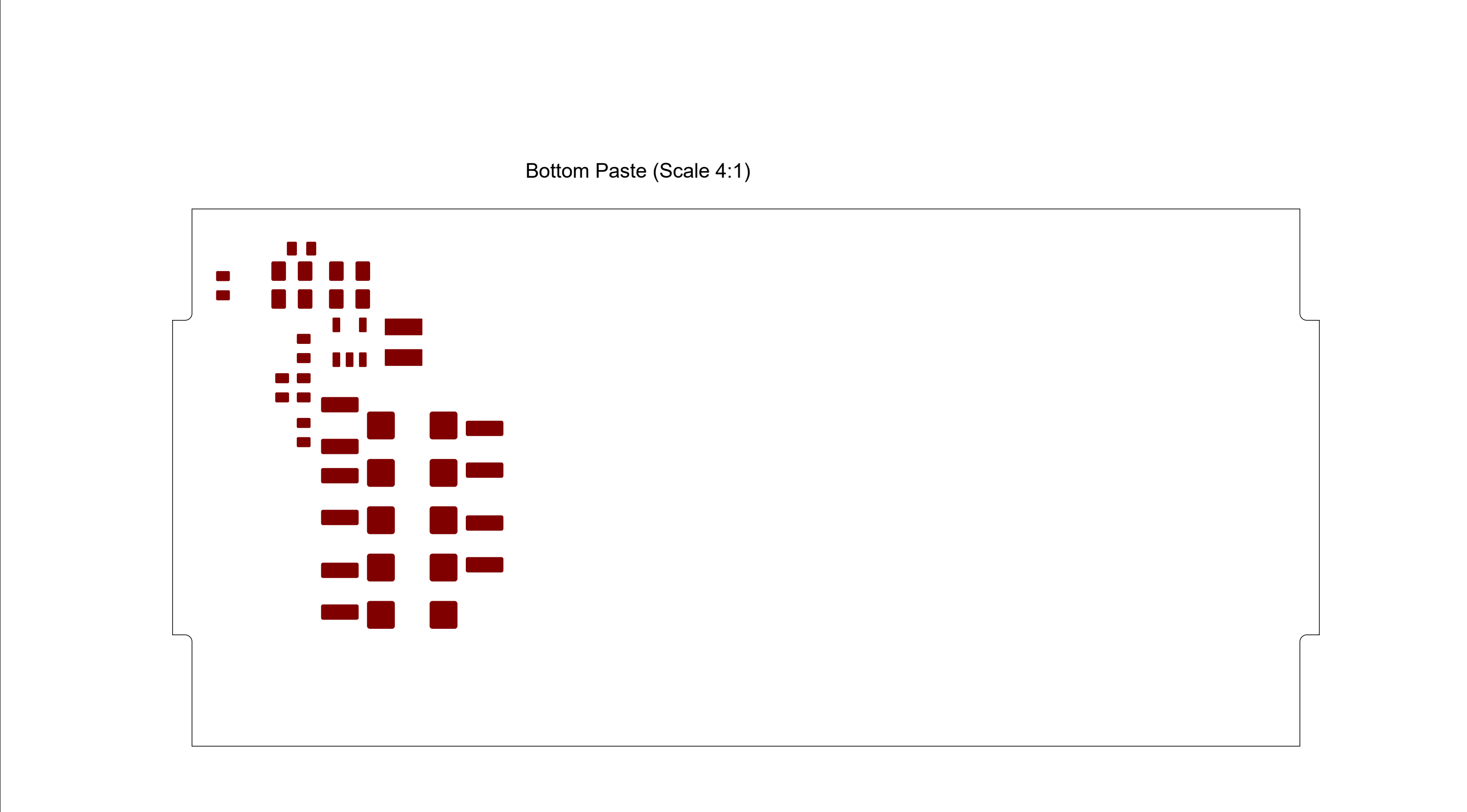


View from Bottom side (Scale 4:1)

123456

PASTE MASK TOP



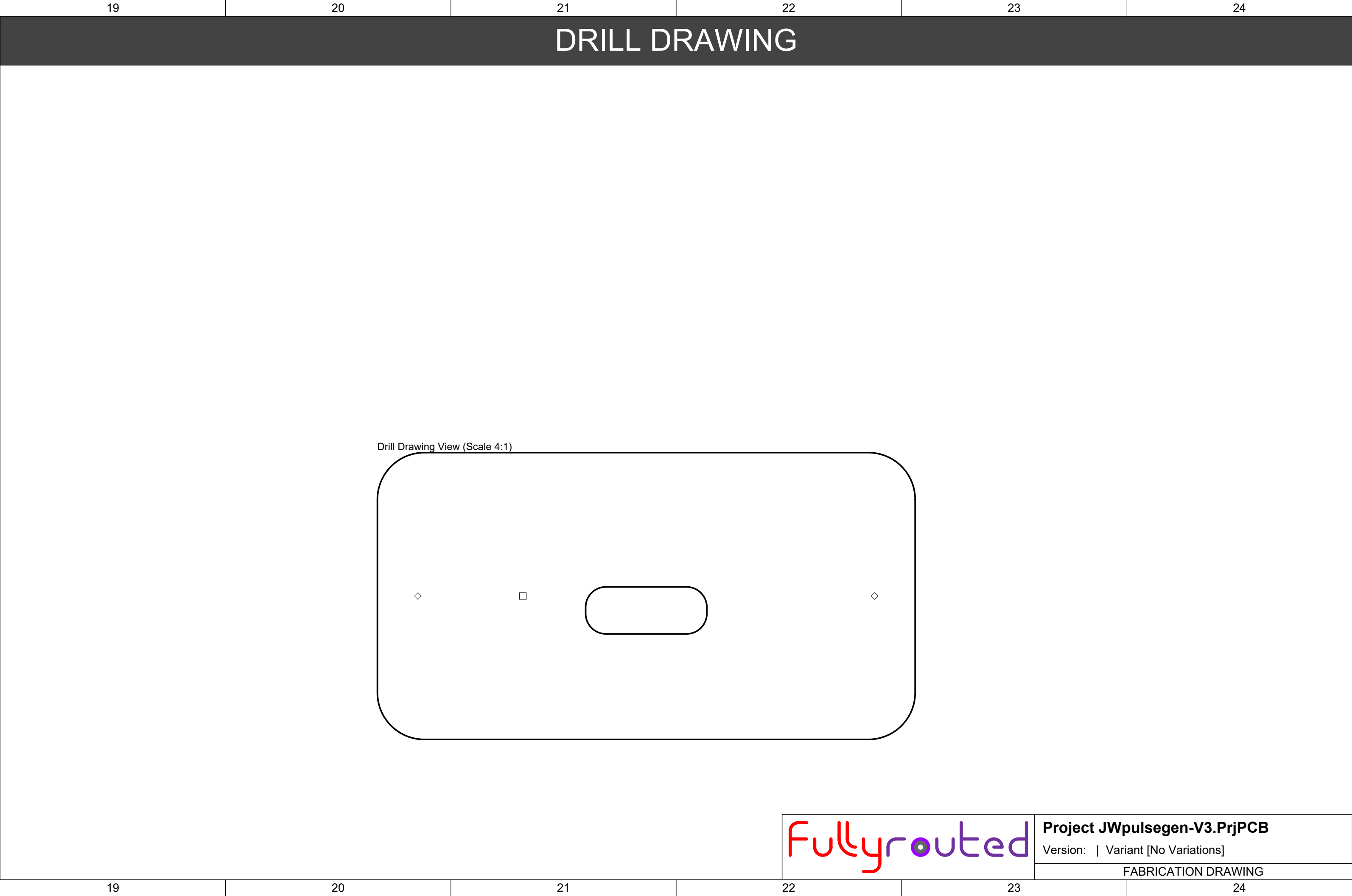


1	2	3	4	5	6
GENERAL					
A	GENERAL				
	1. DO NOT ALTER SUPPLIED COPPER OR DRILL DATA				
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	MATERIALS				
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	10. SOLDER MASK COLOR : BLACK				
C	11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm				
	12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL				
	13. THICKNESS LISTED IN LAYER STACK LEGEND REPRESENT FINAL PRESSED VALUES FOR THE PREPREG				
	14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
	QA, ELECTRICAL TEST AND MARKINGS				
D	15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
				Fullyrouted Project JWpulsegen-V3.PrjPCB Version: Variant [No Variations] FABRICATION DRAWING	
1	2	3	4	5	6

LAYER STACK

Layer Stack Legend

Material	Layer	Thickness	Dielectric Material	Type	Gerber	Dk	Weight	Constructions	Df	Resin
	Top Overlay			Legend	GTO					
Surface Material	Top Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GTS	3.5				
Copper	Top Layer	0.036mm(1.400mil)		Signal	GTL		1oz			
		1.500mm(59.055mil)	FR-4	Dielectric		4.8				
Copper	Bottom Layer	0.036mm(1.400mil)		Signal	GBL		1oz			
Surface Material	Bottom Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GBS	3.5				
	Bottom Overlay			Legend	GBO					
Total thickness: 1.591mm(62.655mil)										



Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

COMPOSITE VIEW FRONT

A

A

B

B

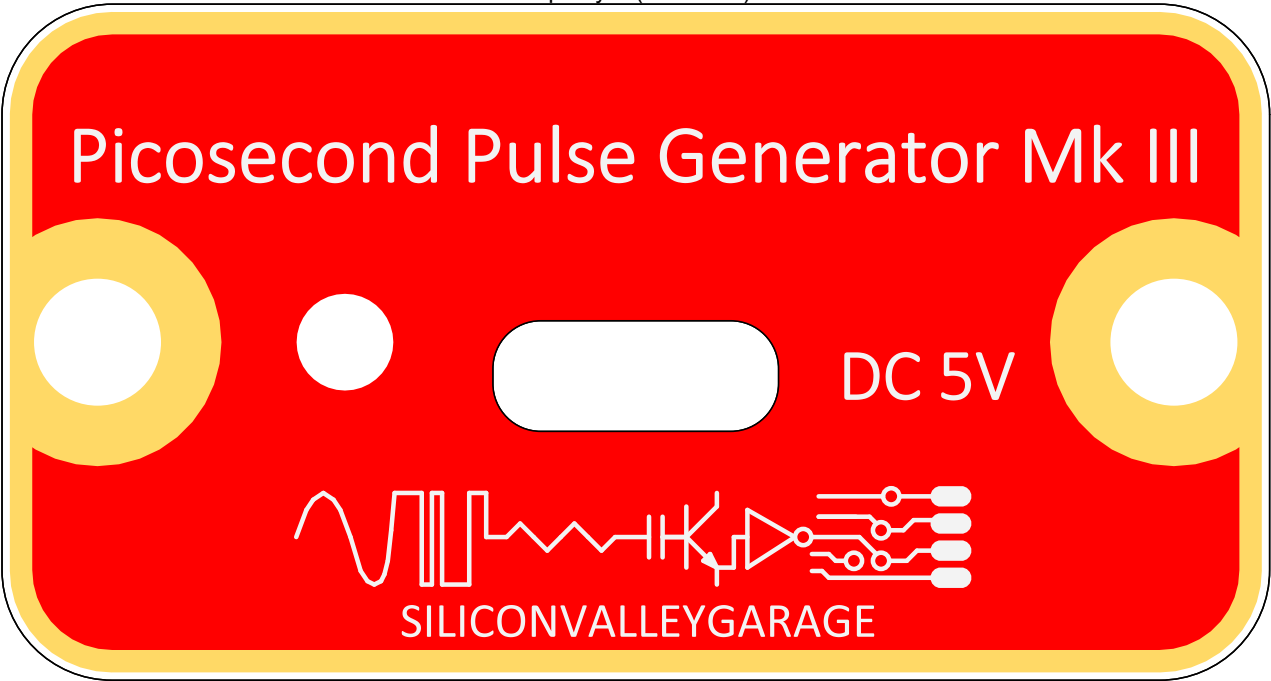
C

C

D

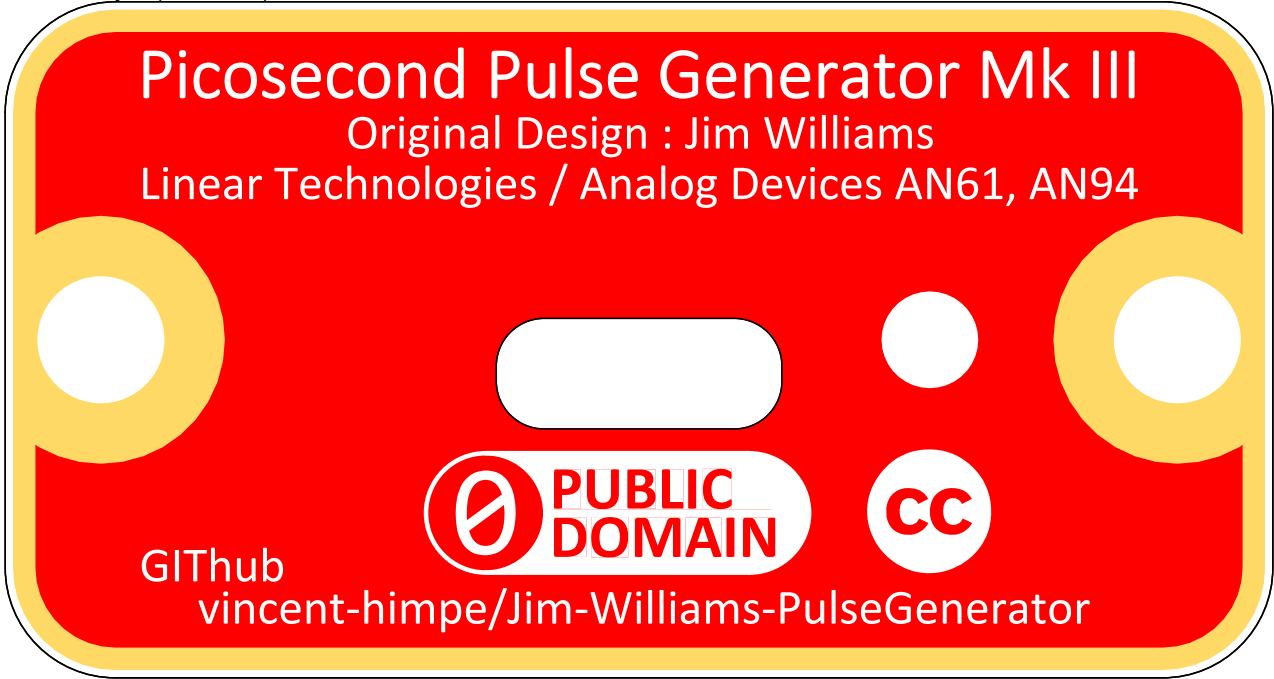
D

Top Layer (Scale 4:1)

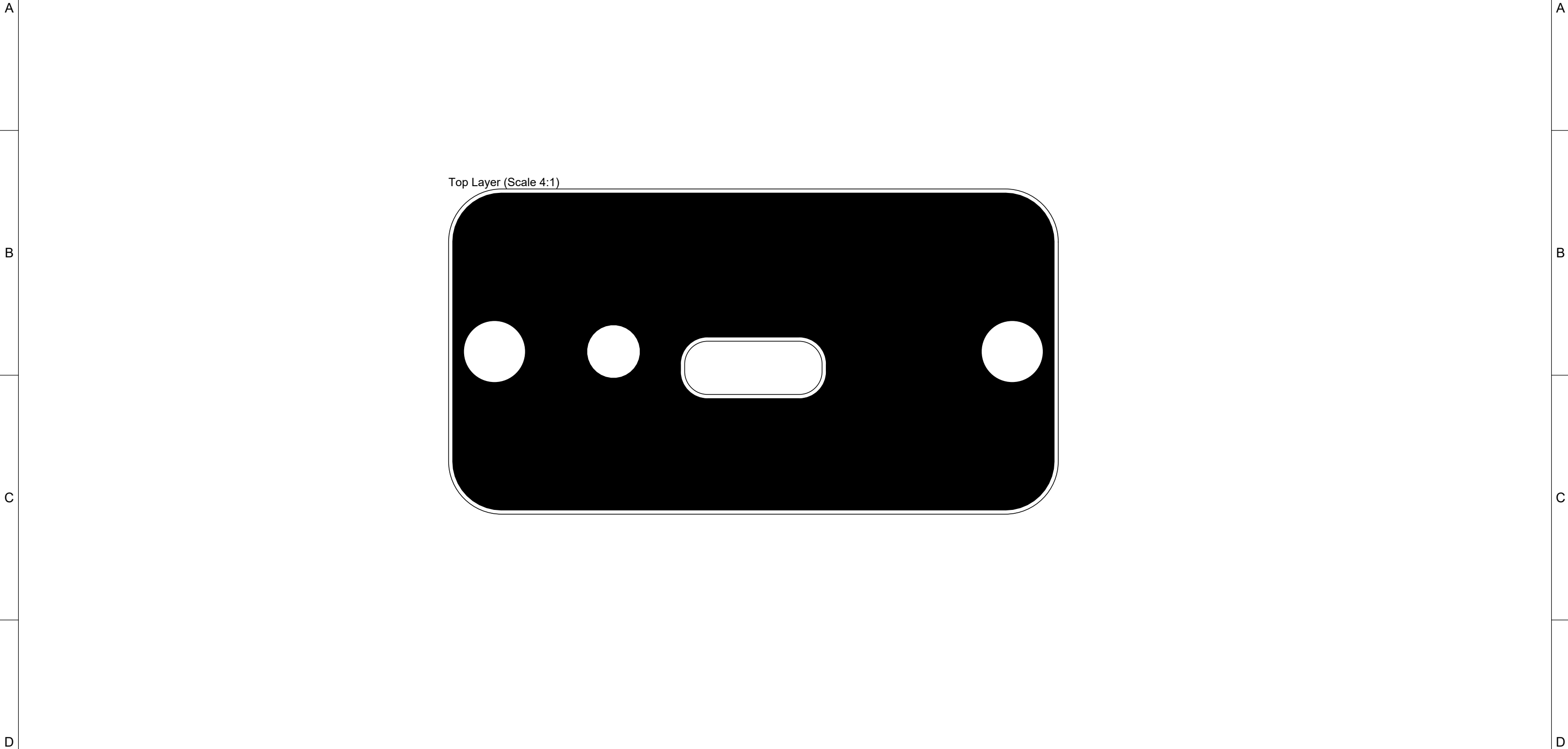


COMPOSITE VIEW BACK

Bottom Layer (Scale 4:1)



LAYER VIEW : TOP LAYER



LAYER VIEW : BOTTOM LAYER

A

A

B

B

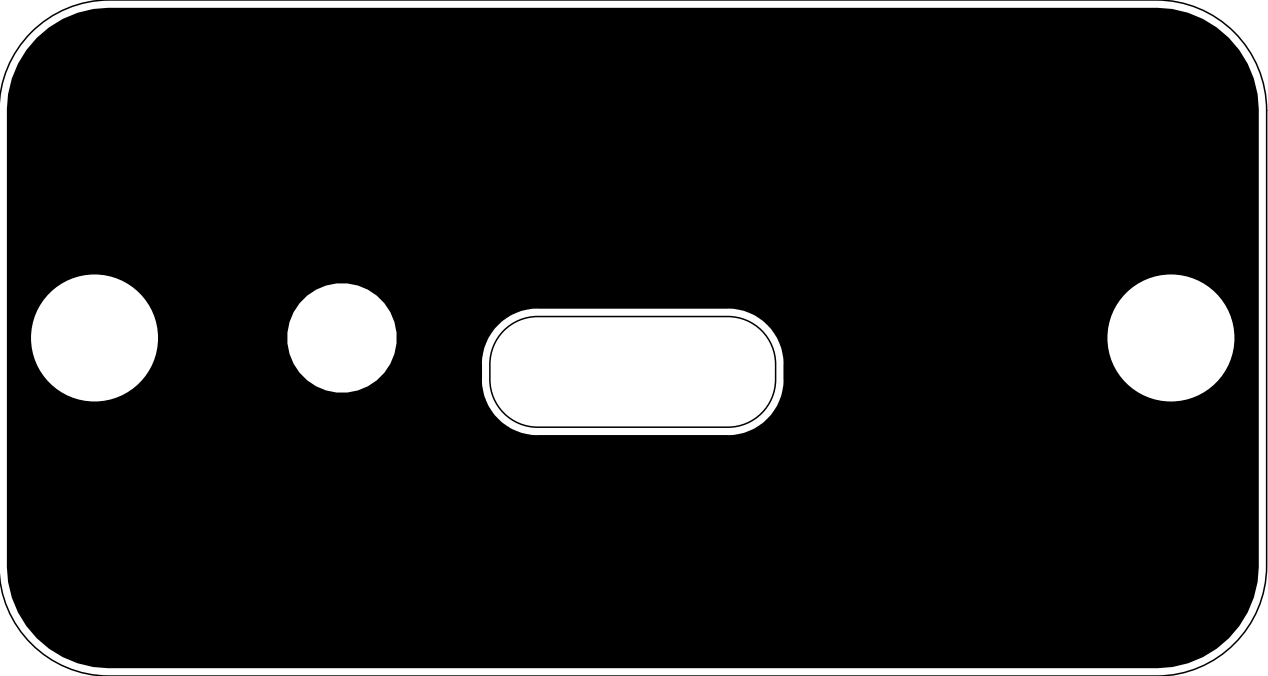
C

C

D

D

Bottom Layer (Scale 4:1)



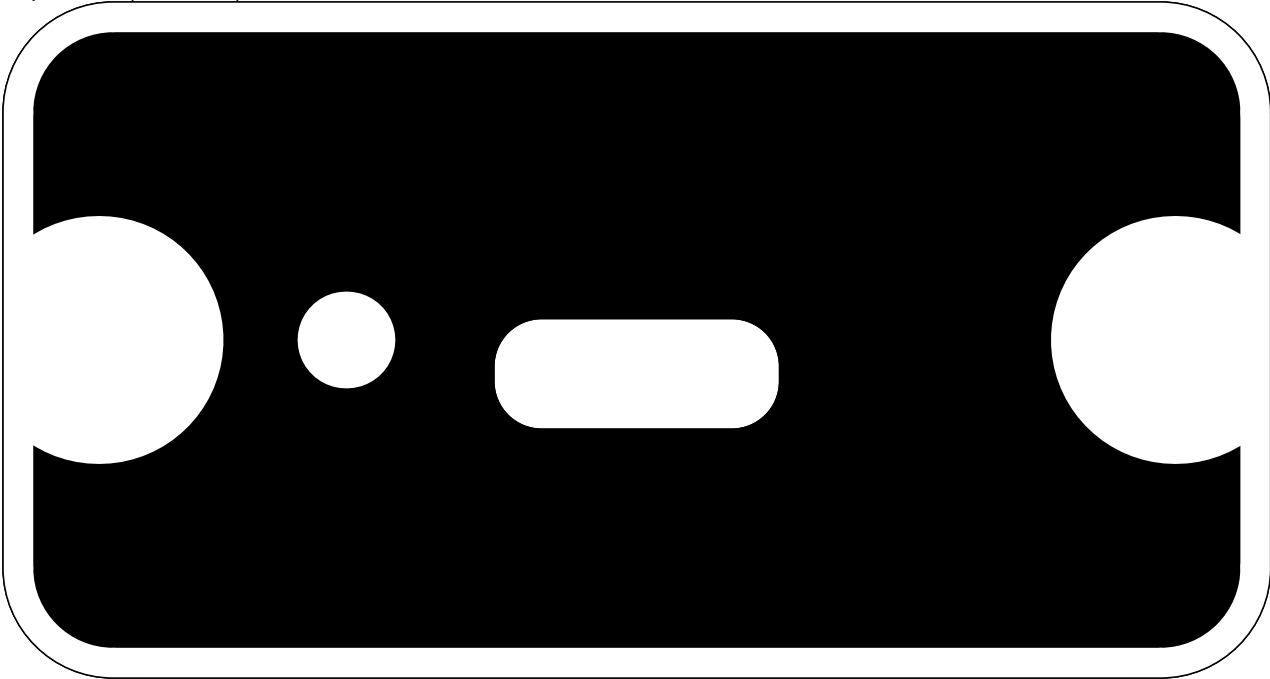
Project JWpulsegen-V3.PrjPCB

Version: | Variant [No Variations]

FABRICATION DRAWING

LAYER VIEW : TOP SOLDER MASK

Top Solder (Scale 4:1)



LAYER VIEW : BOTTOM SOLDER MASK

A

A

B

B

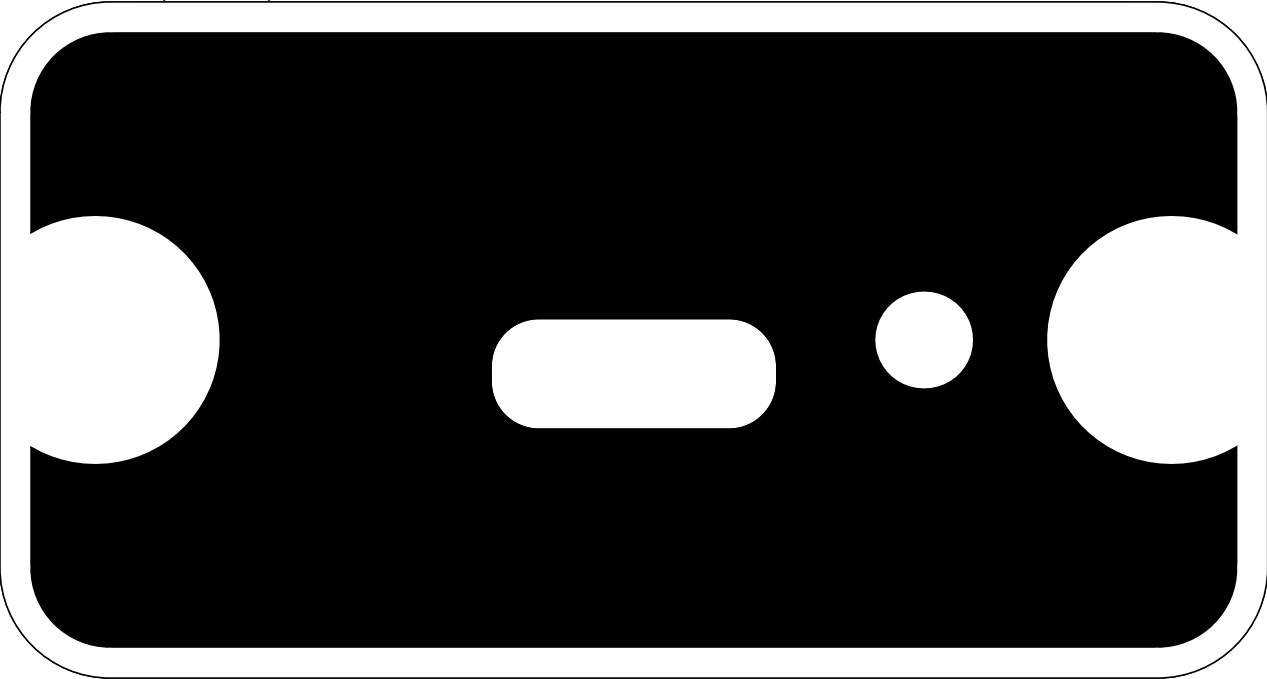
C

C

D

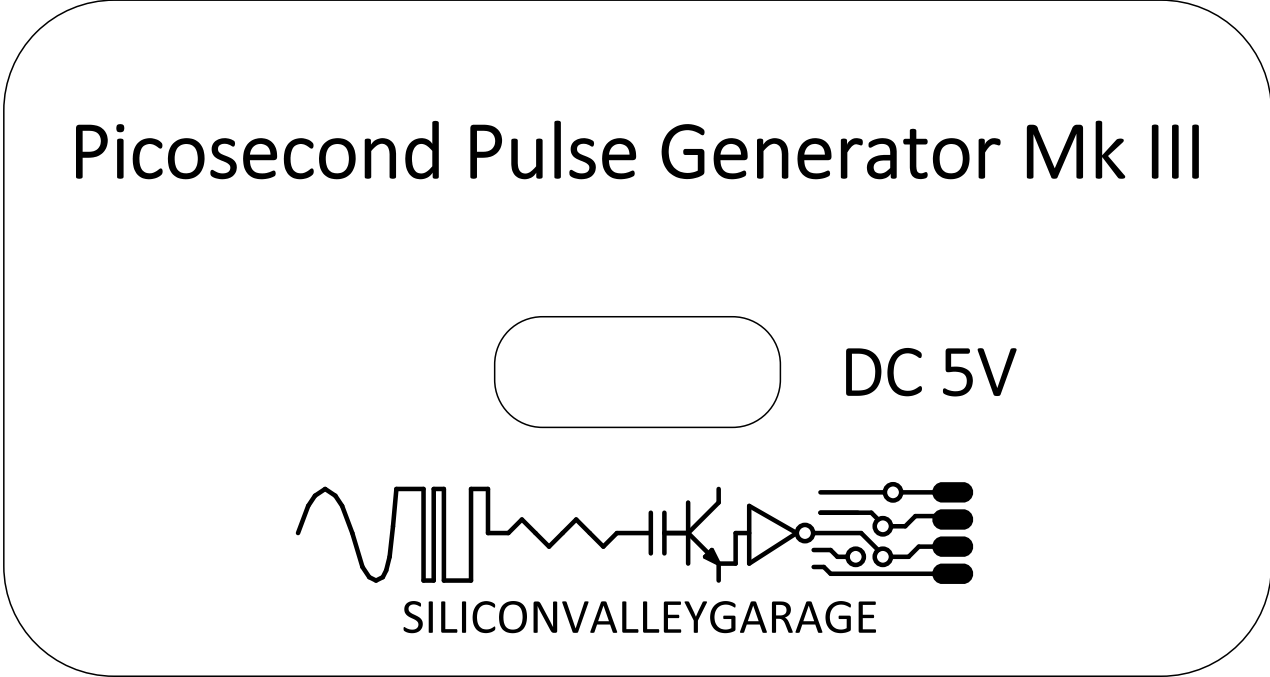
D

Bottom Solder (Scale 4:1)



LAYER VIEW : TOP SILKSCREEN (LEGEND)

Top Overlay (Scale 4:1)



LAYER VIEW : BOTTOM SILKSCREEN (LEGEND)

Bottom Overlay (Scale 4:1)

Picosecond Pulse Generator Mk III

Original Design : Jim Williams

Linear Technologies / Analog Devices AN61, AN94



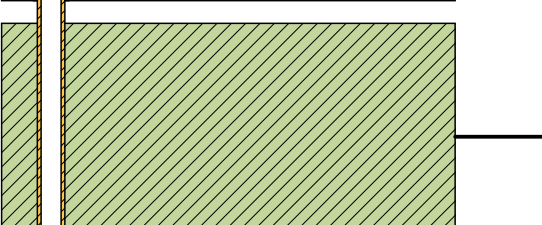
GIThub

vincent-himpe/Jim-Williams-PulseGenerator

1	2	3	4	5	6
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C	11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm				
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				Fullyrouted Project JWpulsegen-V3.PrjPCB Version: Variant [No Variations] FABRICATION DRAWING	
1	2	3	4	5	6

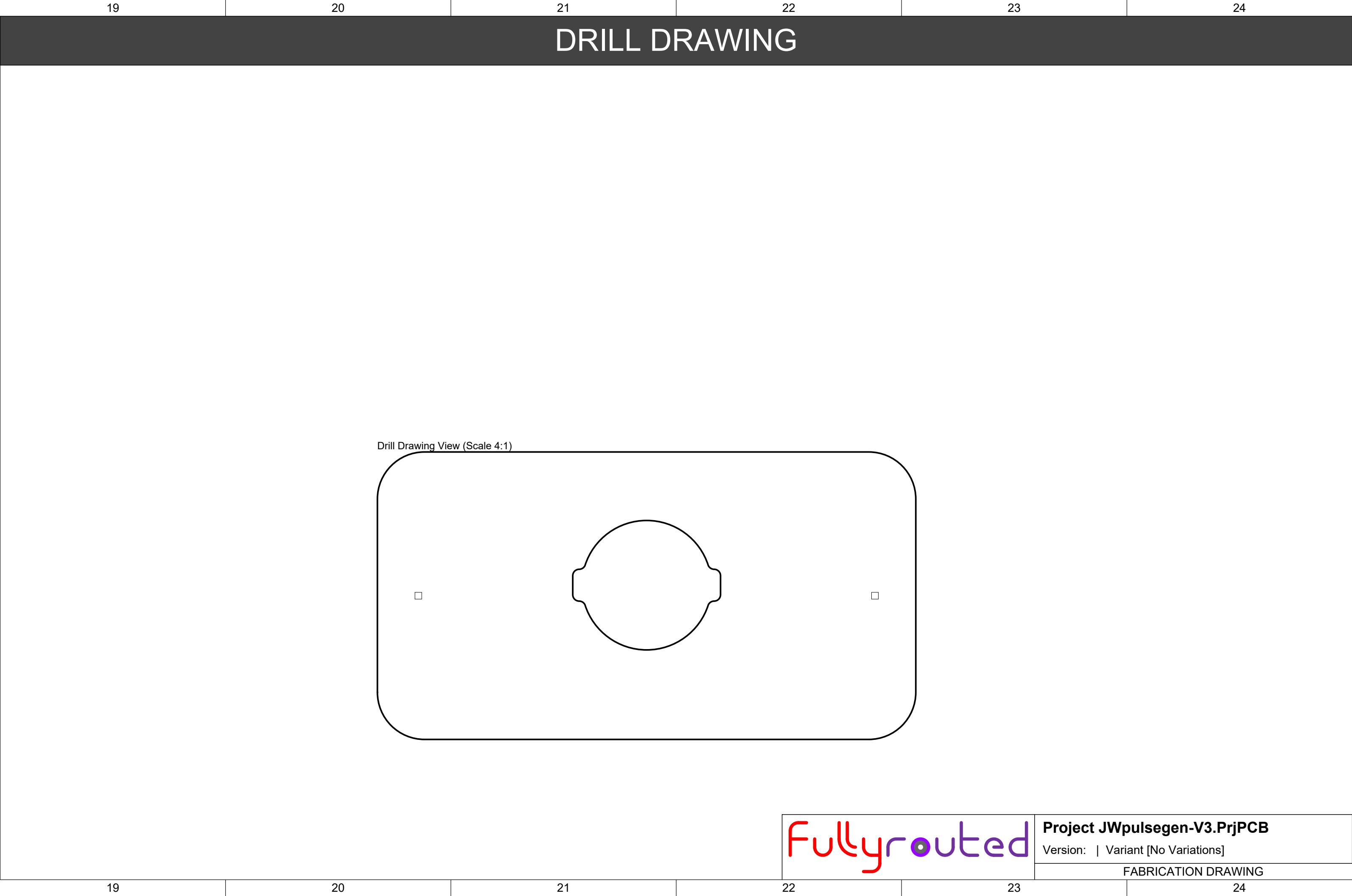
LAYER STACK

Layer Stack Legend

	Material	Layer	Thickness	Dielectric Material	Type	Gerber	Dk	Weight	Constructions	Df	Resin
		Top Overlay			Legend	GTO					
	Surface Material	Top Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GTS	3.5				
	Copper	Top Layer	0.036mm(1.400mil)		Signal	GTL		1oz			
			<i>1.500mm(59.055mil)</i>	<i>FR-4</i>	<i>Dielectric</i>		<i>4.8</i>				
	Copper	Bottom Layer	0.036mm(1.400mil)		Signal	GBL		1oz			
	Surface Material	Bottom Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GBS	3.5				
		Bottom Overlay			Legend	GBO					
Total thickness: 1.591mm(62.655mil)											

Drill Table

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Via / Pad	Pad Shape	Description	Hole Tolerance	Via Type	Via Feature
□	2	4.200mm(165.354mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	2 Total										



COMPOSITE VIEW FRONT

A

A

B

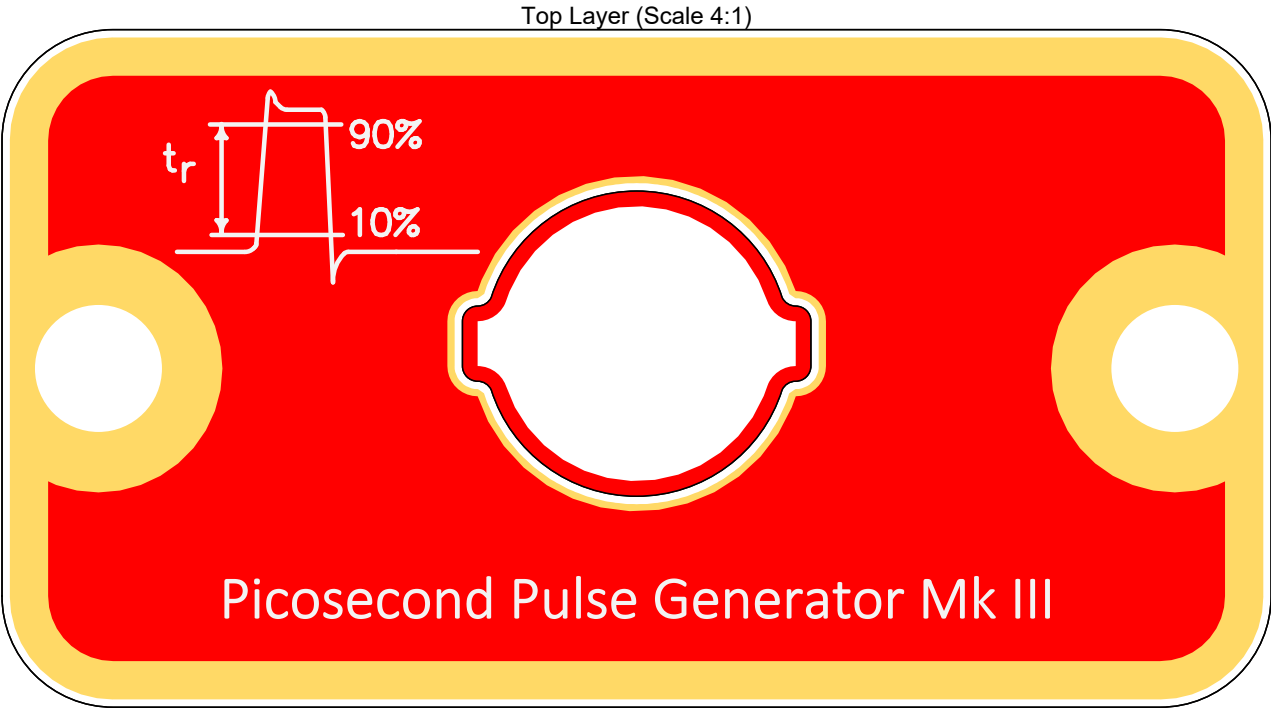
B

C

C

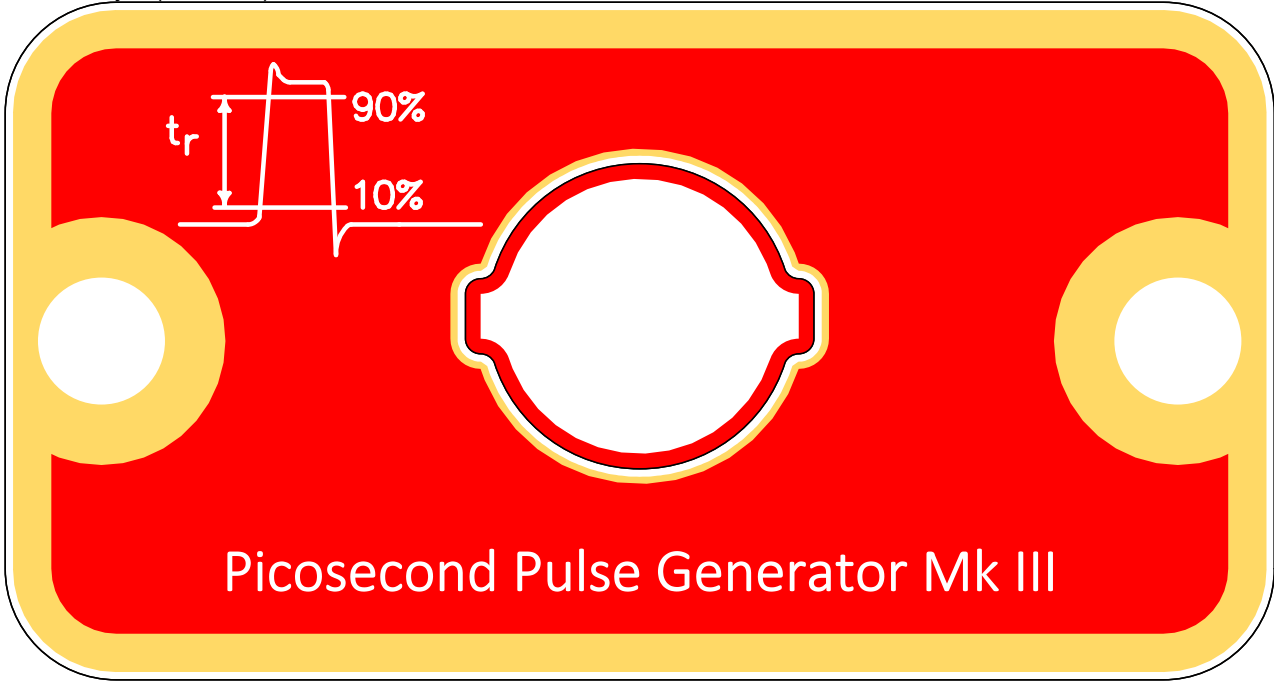
D

D



COMPOSITE VIEW BACK

Bottom Layer (Scale 4:1)



LAYER VIEW : TOP LAYER

A

A

B

B

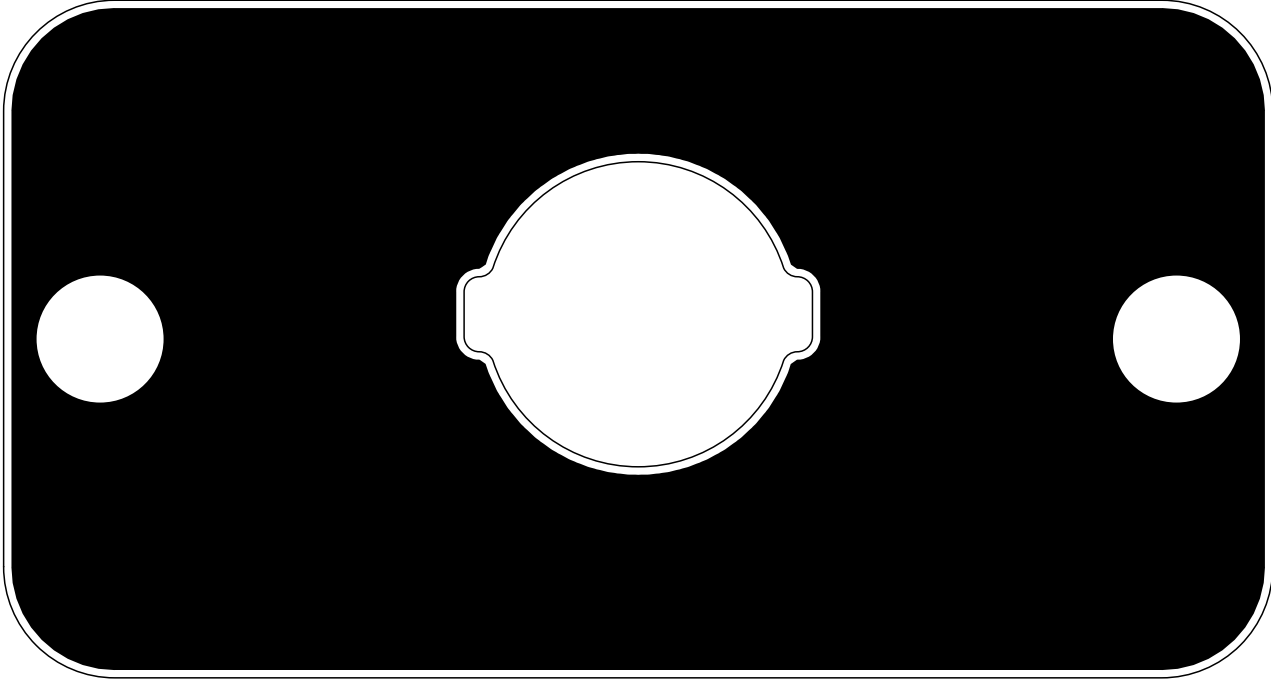
C

C

D

D

Top Layer (Scale 4:1)



LAYER VIEW : BOTTOM LAYER

A

A

B

B

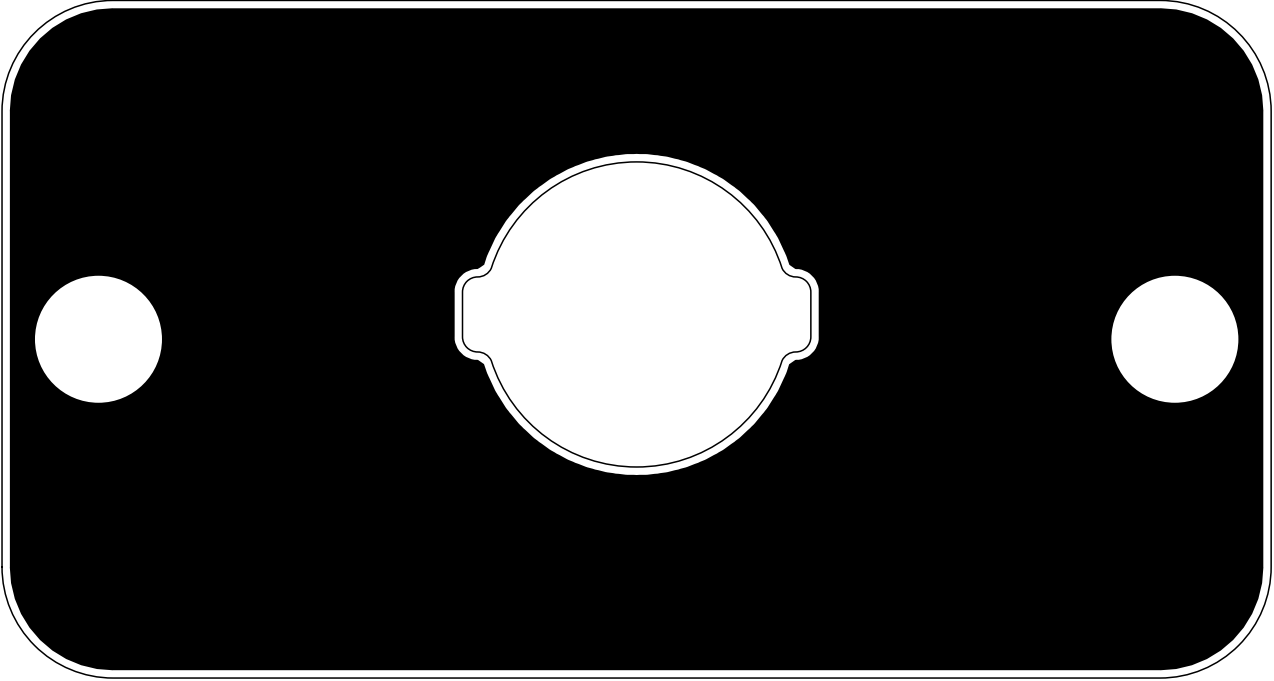
C

C

D

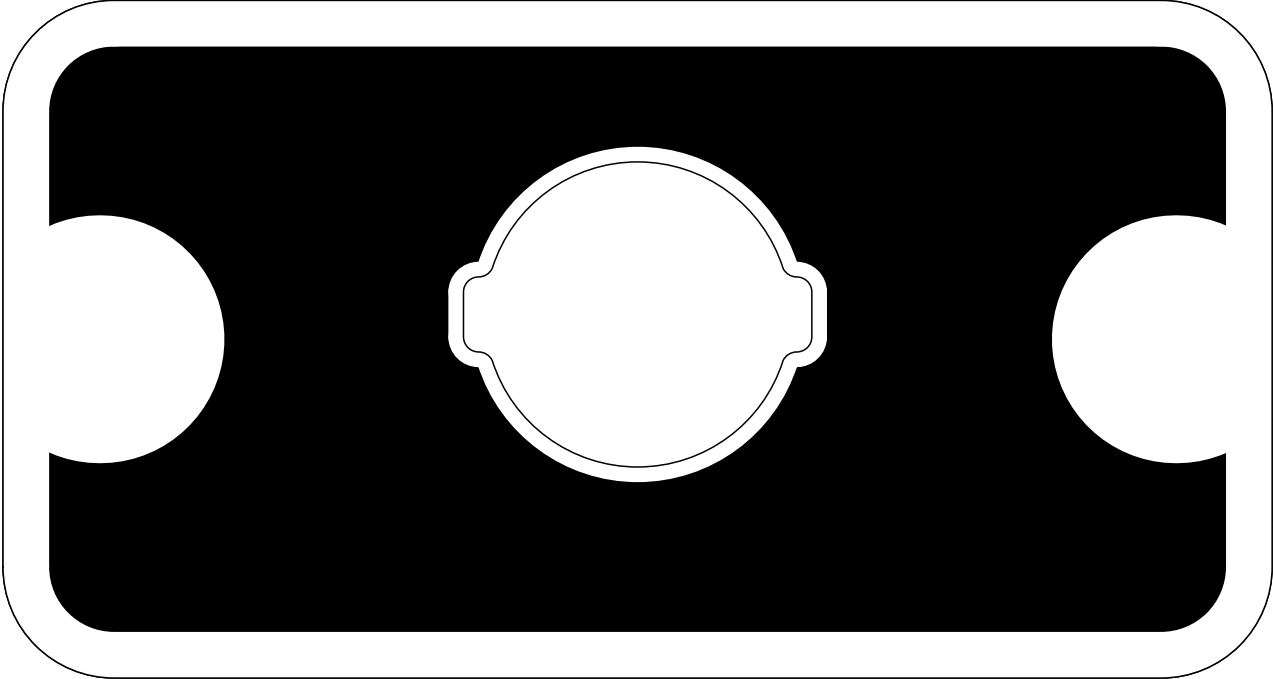
D

Bottom Layer (Scale 4:1)



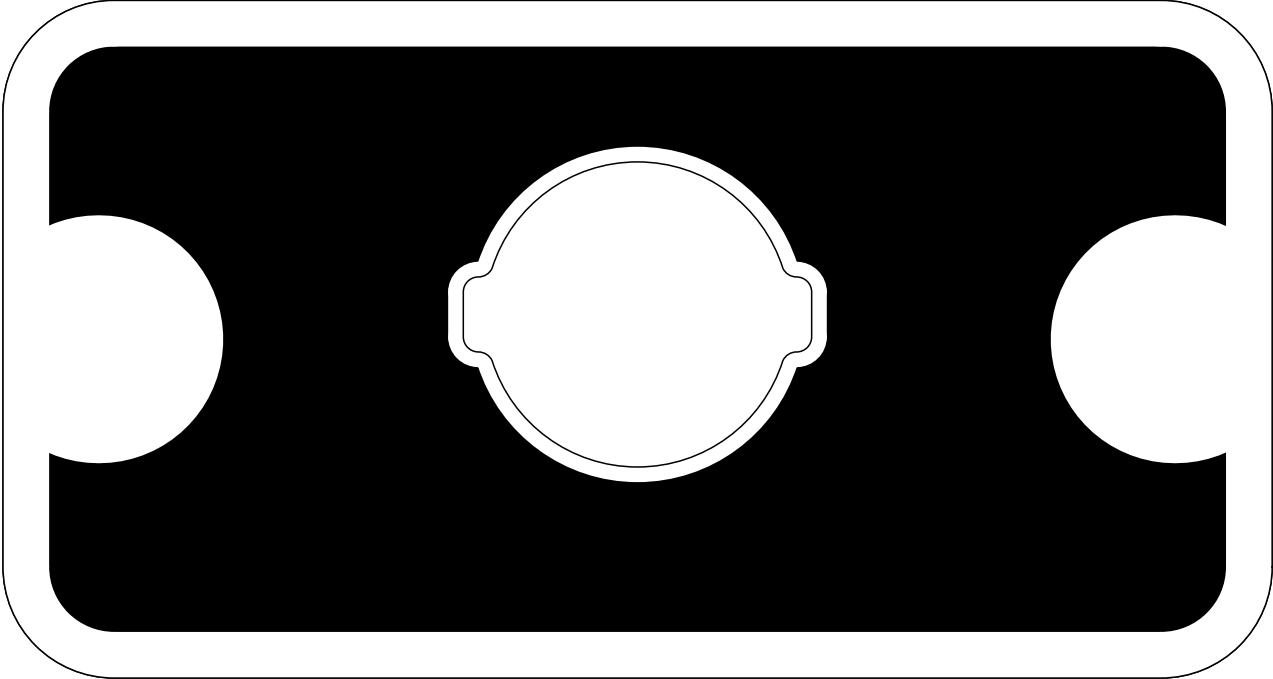
LAYER VIEW : TOP SOLDER MASK

Top Solder (Scale 4:1)

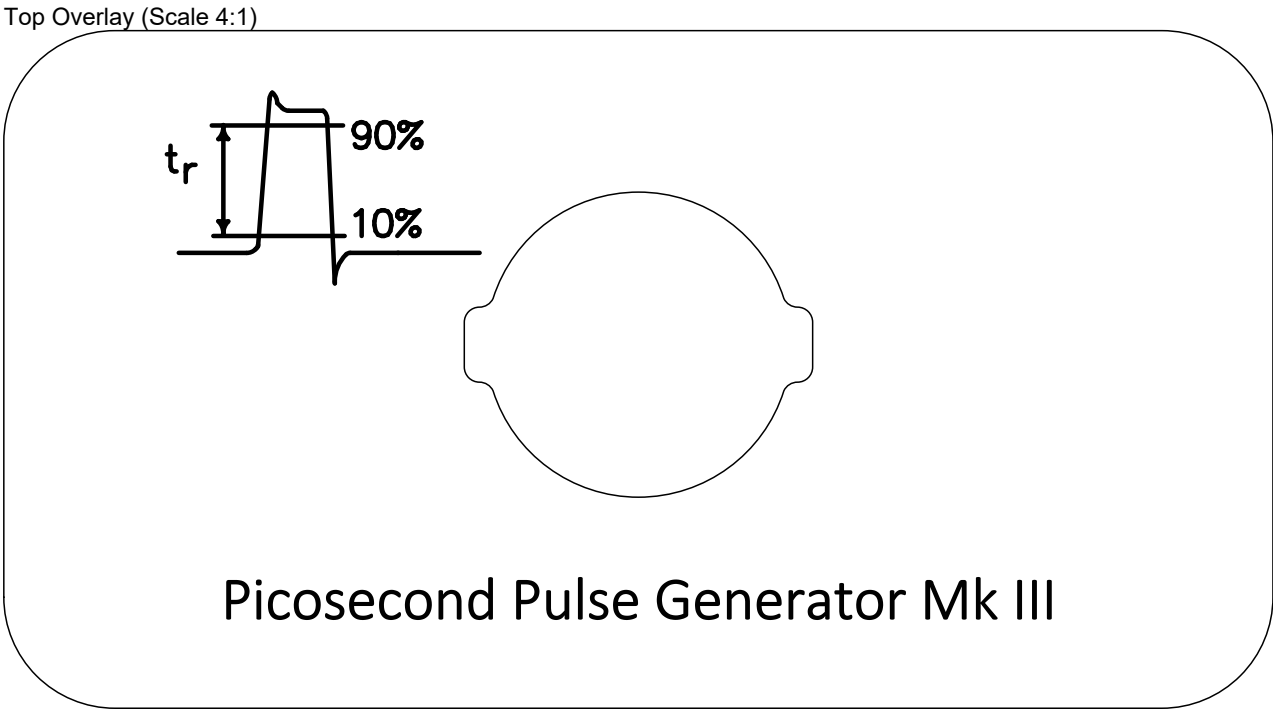


LAYER VIEW : BOTTOM SOLDER MASK

Bottom Solder (Scale 4:1)



LAYER VIEW : TOP SILKSCREEN (LEGEND)



LAYER VIEW : BOTTOM SILKSCREEN (LEGEND)

